Watershed Cosmic Void Finder for Weak Gravitational Lensing Voids on 2D Full-Sky Convergence Maps



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Kosmischer-Leerraum-Sucher basierend auf Wasserscheidentransformation angewandt auf zweidimensionale schwache Gravitationslinsenkonvergenzkarten des gesamten Himmels



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Abstract

Cosmic voids are a major component of the large-scale-structure. A great deal of cosmological information can be extracted from cosmic voids. Normally, voids are identified from galaxy distributions, which are called galaxy voids. Previous works have proved that 2D weak lensing voids are also a good cosmological probe. As more and more high accuracy surveys (*Euclid*, *DES*...) projects have been put into service, weak lensing becomes a powerful cosmological probe. There are many methods for the identifications of weak lensing voids, the watershed algorithm is a popular parameter free topological method for identifying voids. We have developed and tested a new full-sky watershed voids finder (**2DWVF**) with the *Takahashi* simulation. We measured and analysed a range of void statistics, including the void size distribution, the ellipticity distribution, and the stacked convergence profiles. We also ran the WVF with the reconstructed weak lensing maps of the *CosmogridV*1 simulation to produce forecasts of cosmological parameter constraints with our **2DWVF**. The void size and ellipticity distributions were used to train a Gaussian process regression emulator. This emulator was used to generate likelihood contours and provide parameter constraints from mock observations. We also applied this void finder to real data from *DES* successfully, proving that the **2DWVF** is compatible not only for simulation but also real observation.

Keywords **:** cosmology, large-scale structure of universe, weak gravitational lensing

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1 Introduction

In modern cosmology the distribution of matter on Megaparsec scales is not homogeneous, the anisotropic gravitational collapse leads to the evolution of the large scale structure (LSS) (Zel'dovich, 1970) and this foam-like structure is called the cosmic web. Fig.(1) shows the LSS generated from the Millennium-Simulation, it consists of filaments, walls, interconnecting nodes and voids. The filaments and walls are the largest known structures in the Universe, these structures accrete matter from surrounding space. The points at which filaments and walls cross are nodes, which correspond to the largest over-densities in the Cosmic web, Galaxy clusters. In stark contrast to this, the vast and relatively empty spaces between filaments are called cosmic voids, where the voids dominate most volume of the Universe (Platen et al., 2007).

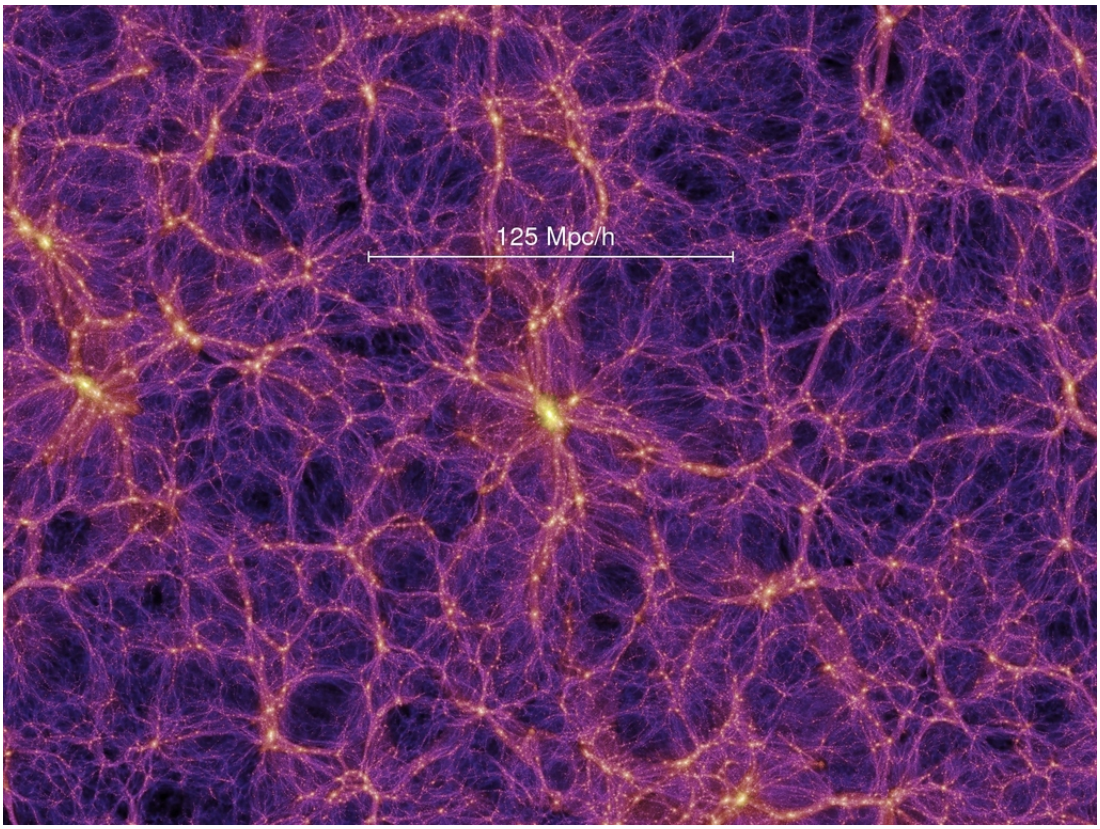


Figure 1: The LSS generated in Millennium-Simulation (Springel et al., 2005)

Cosmic voids are useful for studying different cosmological models, because the formation of cosmic voids depends on underlying cosmology. Different cosmological parameters result in different properties of voids, which can be revealed by the shape and size distributions as well as the overdensity profiles (Hamaus et al., 2020). Many cosmological models have been tested with void statistics, such as dark energy (Lavaux and Wandelt, 2012, Biswas et al., 2010, Bos et al., 2012, Lee and Park, 2009), modified gravity (Davies et al., 2019, Clampitt et al., 2013, Cai et al., 2015, Perico et al., 2019), the massive neutrinos (Massara et al., 2015, Schuster et al., 2019) and the primordial non-Gaussianity

(Chan et al., 2019, Kamionkowski et al., 2009).

In most cases, voids are identified in the 3D matter distribution (Sutter et al., 2014), where galaxies are used as tracers of the underlying matter distribution. Galaxy voids are widely used, where voids are identified as underdense regions in the galaxy distribution (Cautun et al., 2016, Hamaus et al., 2020). But 2D weak gravitational lensing voids have also been shown to be a useful cosmological probe recently (Davies et al., 2019).

As light travels through the Universe, it is deflected when it passes through gravitational potentials. This results in images of distant sources that are distorted by the foreground matter. This phenomenon is called gravitational lensing, which is generally split into three regimes, strong lensing, weak lensing, and micro lensing. In the strong lensing regime the distortion is large and can form Einstein rings, arcs and multiple images. In the micro lensing regime there is no shape distortion, only the amount of light received from the background object changes in time. In the weak lensing (WL) regime, this effect is smaller than strong lensing but larger than micro lensing, which is the result of the gravitational lensing induced by the total matter distribution along the line-of-sight. WL can be measured through the correlations in distortions of galaxy images. WL is sensitive to the total matter distribution, which makes WL a observable tool to probe the total matter distribution of the LSS in the Universe (Bartelmann and Schneider, 2001). With the WL convergence (κ) field we can identify 2D WL voids, which generally have higher signal-to-noise-ratio (SNR) (Davies et al., 2020). Furthermore, the Galaxy - Halo connection is not fully understood, making the nature of the galaxy bias a challenge for traditional void studies. Davies et al. (2019) have proved the effectiveness of WL voids statistics in the context of constraining a class of modified gravity models, where the models can be constrained better with 2D WL voids than with 3D galaxy voids.

Unlike the Galaxy clusters, there is no universal definition for a void. They are generally defined as extended underdense regions, but this broad definition allows for a range of possible numerical algorithms that fulfil this criteria. There are several methods for the identification of WL voids, which include:

- WL minima: The local minima in the κ field can be considered as WL voids. In Chang et al. (2018) the deepest minima were shown to correspond to supervoids along the line-of-sight.
- Troughs: Voids can be identified by randomly placing circles with fixed size in κ field and discarding the circles that have an overdensity of tracers, only leaving that have the least tracers.
- Tunnels: The voids are identified as the largest circles that do not enclose tracers (Cautun et al., 2018).
- Spherical void finder (SVF): The voids are defined as the largest circles whose mean κ is below the given threshold.
- Watershed void finder (WVF): The voids are identified as the watershed in the WL

maps, the nodes, filaments and walls would then correspond to the boundary of the voids.

In all of the void finders listed above, only the WVF can extract the morphology and shape information of the voids. Furthermore, Davies et al. (2020) have studied and compared different WL void finders. Their results showed that the Minima void finder has the highest SNR and the WVF has the second highest SNR. Additionally the WVF was shown to be less sensitive to contamination from noise compared to the WL Minima. These features make the WVF a promising WL void definition that may be complementary to other WL void finders.

Previous versions of the WVF have only been applied to small flat patches of the sky. In this thesis we develop a code **2DWVF** that allows the WVF to be run directly on Healpix maps, (a popular format for astronomical data) which represent the full curved sky. We use the **2DWVF** to produce WL void catalogues from the convergence maps, and we use these catalogues to measure a range of void summary statistics. This includes the void size and ellipticity distribution, the statistical tool for κ profile based on two different methods, which one is the centroid method and the other is the boundary method (Cautun et al., 2016). We use the Takahashi simulations to test and develop our algorithm (Takahashi et al., 2017).

This paper is organized as follows: in Chapter 2 we describe the relevant weak lensing theory. In Chapter 3 we present employed the void-finding and statistical algorithms. The results from the void finder are described in Chapter 4. The void abundance, void ellipticity and void convergence profile are also presented in the same chapter. We also tested our algorithms with the data of DES and CosmoGridV1, the results and cosmological constraints are presented in Chapter 5. Finally, we discuss and summarize our results in Chapter 6.

2 Theory

In this chapter we present the theory behind weak lensing and its statistics.

2.1 Basic theory of weak gravitational lensing

First, we show a schematic description of gravitational lensing in Fig.(2):

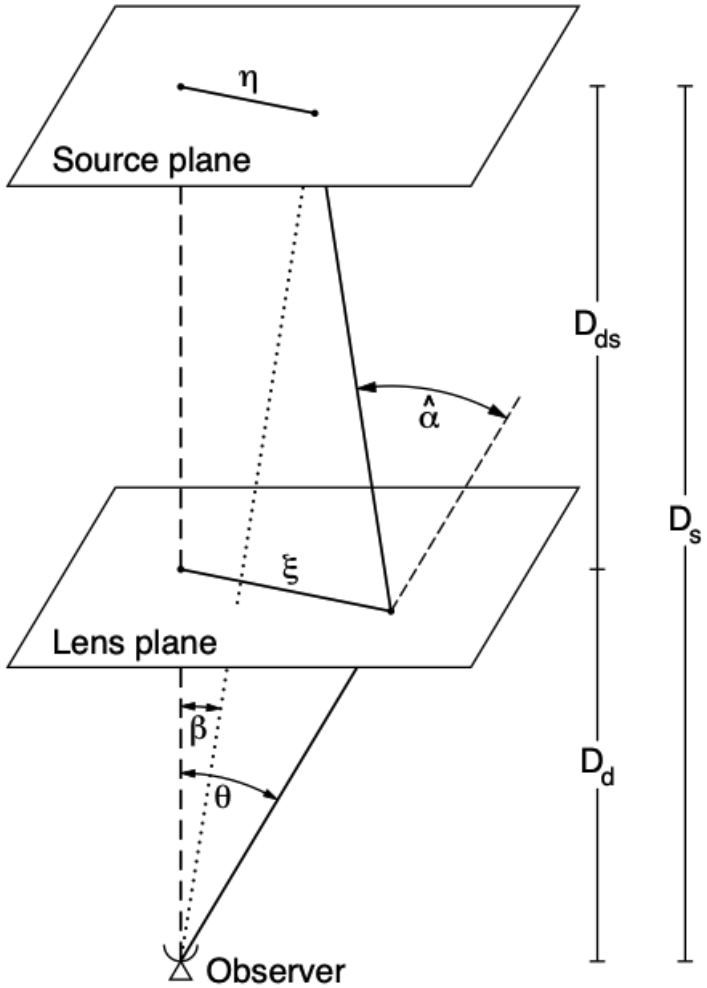


Figure 2: Illustration of a lensing system (Bartelmann and Schneider, 2001).

where D_s is the distance between the source and the observer, D_d is the distance between the lens and the source, D_{ds} is the distance between the lens and the observer. The light is emitted by a source which has a distance η from the central optical axis and propagates parallel to it, the light ray is deflected by an angle $\hat{\alpha}$ and reaches the observer. The resulting image would have an angular separation θ from the optical axis in the sky which is different from the actual separation of the source β.

Now we need an equation to relate the true position of the source to its observed position in the sky, the lens equation of a lensed image is given by:

$$\boldsymbol{\eta} = \frac{D_s}{D_d}\boldsymbol{\xi} - D_{ds}\hat{\boldsymbol{\alpha}}(\boldsymbol{\xi}). \tag{1}$$

With the angular coordinates $\boldsymbol{\eta} = D_s\boldsymbol{\beta}$ and $\boldsymbol{\xi} = D_d\boldsymbol{\theta}$, we can transform Eq.(1) to

$$\boldsymbol{\beta} = \boldsymbol{\theta} - \frac{D_{ds}}{D_s}\hat{\boldsymbol{\alpha}}(D_d\boldsymbol{\theta}) = \boldsymbol{\theta} - \boldsymbol{\alpha}(\boldsymbol{\theta}). \tag{2}$$

The interpretation of this equation is that the true position $\boldsymbol{\beta}$ can be seen by the observer at angular positions $\boldsymbol{\theta}$. If a source is much smaller than the angular scale on which the lens properties change, the deformation of the lensed image with respect to its unlensed image can be represented by a Jacobian matrix A:

$$A_{ij} = \frac{\partial\beta_i}{\partial\theta_j} = \delta_{ij} - \frac{\partial\alpha_i}{\partial\theta_j}. \tag{3}$$

By neglecting lens-lens coupling, with the Born approximation, the deflection angle can be expressed as the gradient of 2D lensing potential ψ, which is given by

$$\psi(\boldsymbol{\theta}, \chi) = \frac{2}{c^2}\int_0^{\chi}\frac{\chi - \chi'}{\chi\chi'}\Phi(\chi'\boldsymbol{\theta}, \boldsymbol{\theta})\mathrm{d}\chi', \tag{4}$$

where ψ is the comoving distance to the source, the integration variable χ' is the comoving distance from the observer to the lens, Φ is the 3D lensing potential of the lens and c is the speed of light. Φ is related to the non-relativistic matter density contrast $\delta = \rho/\overline{\rho} - 1$, the relationship can be expressed by the Poisson equation

$$\nabla^2\Phi = 4\pi\mathrm{G}a^2\overline{\rho}\delta, \tag{5}$$

where G is the gravitational constant, a is the scale factor and $\overline{\rho}$ is the mean matter density of the Universe.

In Eq.(4) we can find that the lensing potential ψ is an integral of the matter distribution along the line of sight from the source to the observer.

With $\boldsymbol{\alpha} = \boldsymbol{\nabla}\psi$, Eq.(3) can be expressed in terms of ψ

$$A_{ij} = \frac{\partial\beta_i}{\partial\theta_j} = \delta_{ij} - \partial_i\partial_j\psi, \tag{6}$$

where partial derivatives are taken with respect to θ. The matrix $\boldsymbol{A}$ can be parameterized in terms of convergence κ and shear $\boldsymbol{\gamma} = \gamma_1 + \mathrm{i}\gamma_2 = \gamma e^{2\mathrm{i}\phi}$. ϕ is the orientation angle of

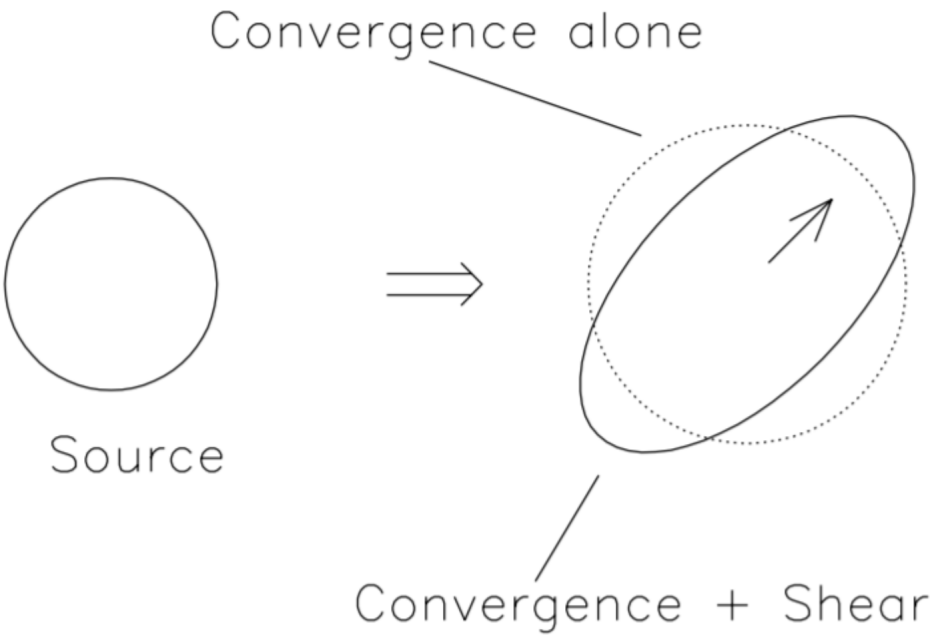


Figure 3: Effects of lensing distortion (Narayan and Bartelmann, 1997).

the shear.

$$\boldsymbol{A} = \begin{pmatrix} 1-\kappa-\gamma_1 & -\gamma_2 \\ -\gamma_2 & 1-\kappa+\gamma_1 \end{pmatrix} = (1-\kappa)\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} - \gamma \begin{pmatrix} cos(2\phi) & sin(2\phi) \\ sin(2\phi) & -cos(2\phi) \end{pmatrix}. \tag{7}$$

Fig.(3) shows how gravitational lensing distorts the image of a source galaxy whose shape is intrinsically spherical.

The convergence and shear are related to the lensing potential ψ through the equations:

$$\kappa = \frac{1}{2}\nabla^2_{\boldsymbol{\theta}}\psi \tag{8}$$

$$\begin{cases} \gamma_1 = \frac{1}{2}(\nabla_{\boldsymbol{\theta}_1}\nabla_{\boldsymbol{\theta}_1} - \nabla_{\boldsymbol{\theta}_2}\nabla_{\boldsymbol{\theta}_2})\psi \\ \gamma_2 = \nabla_{\boldsymbol{\theta}_1}\nabla_{\boldsymbol{\theta}_2}\psi \end{cases}, \tag{9}$$

where $\nabla_{\boldsymbol{\theta}} = (\chi')^{-1}\nabla$. We can substitute the lensing potential in Eq.(4) with the expression in Eq.(8), notice that the Laplacian operator can only operate with the Newtonian potential within the integrand because it acts on angular coordinates. By introducing the Poisson equation Eq.(5) in comoving coordinates, the convergence can be expressed in:

$$\kappa(\boldsymbol{\theta}, \chi) = \frac{3H_0^2\Omega_m}{2c^2}\int_0^{\chi} \frac{\chi-\chi'}{\chi}\chi'\frac{\delta(\chi'\boldsymbol{\theta}, \chi')}{a(\chi')}\mathrm{d}\chi', \tag{10}$$

where H_0 is the Hubble parameter at redshift zero, Ω_m is the density parameter of total matter in the Universe, $\delta(\chi\boldsymbol{\theta}, \chi)$ is the matter density perturbation field and $a(\chi)$ is the scale factor. Eq.(10) shows that if a single light source is located at comoving distance χ with angular position $\boldsymbol{\theta}$, then the observed WL convergence can be interpreted as the projected density perturbation weighted by $(\chi - \chi')\chi'/\chi$ (lensing efficiency factor) along the line of sight.

In the derivations above we assume a fixed source plane, just like the Takahashi simulation, the source redshift is fixed. But actually in many other simulations and observations (e.g. DES, CosmoGridV1) the source galaxies don't occupy a single plane at a fixed distance from the observer. In the work of Kilbinger (2015), Eq.(10) is weighted with the galaxy probability distribution in comoving distance, $n_s(\chi)d\chi$:

$$\kappa_{\rm eff}(\boldsymbol{\theta}) = \int_0^{\chi_{max}} n(\chi)\kappa(\boldsymbol{\theta}, \chi)\mathrm{d}\chi \tag{11}$$

$$= \frac{3H_0^2\Omega_m}{2c^2}\int_0^{\chi_{max}} \mathrm{d}\chi g(\chi)\chi\frac{\delta(\chi\boldsymbol{\theta}, \chi)}{a(\chi)}, \tag{12}$$

where χ_{max} is the maximal radial comoving distance and $g(\chi)$ is:

$$g(\chi) = \int_\chi^{\chi_{max}} \mathrm{d}\chi' n_s(\chi')\frac{\chi' - \chi}{\chi'}. \tag{13}$$

From the discussion above we know that both convergence and shear are derived from the second order derivative of the lensing potential, which means the convergence and shear are dependent from each other. The relation can be written down in Fourier space where the 2D angular position $\boldsymbol{\theta}$ transforms to the multipole $\boldsymbol{l}$:

$$\gamma(\boldsymbol{l}) = \frac{(l_x + \mathrm{i}l_y)^2}{l^2}\kappa(\boldsymbol{l}) = e^{2\mathrm{i}\phi_l}\kappa(\boldsymbol{l}), \tag{14}$$

where $l \neq 0$, $l = (l_x^2 + l_y^2)^{\frac{1}{2}}$ and $\phi_l = \arctan(\frac{l_y}{l_x})$ is the polar angle of $\boldsymbol{l}$.

2.2 Statistics of cosmology and weak lensing

In cosmological principle, The Universe is homogeneous and isotropic. However, this can only be held true at large length scales (∼100 Mpc). As the result of the gravitational evolution of the Universe in small scale, there are prominent inhomogeneities in the matter density field upon the homogeneous and isotropic background on.

We use the variance of the density perturbation field $\langle\delta^2\rangle$ to describe inhomogeneities, because the average of the density perturbation field is always zero. But the variance just indicates the strength of the inhomogeneity at single location, the spatial correlations

of density perturbations between different points are not taken into account. Hence we introduce another statistic 2-point correlation function ξ:

$$\xi(\boldsymbol{r}) = \langle \delta(\boldsymbol{x})\delta(\boldsymbol{x}+\boldsymbol{r})\rangle, \tag{15}$$

where $\boldsymbol{r}$ is the displacement vector between two points and the angle bracket indicates an average over the volume. It can be represented by Fourier transform:

$$\xi(\boldsymbol{r}) = \langle \delta(\boldsymbol{x})\delta(\boldsymbol{x}+\boldsymbol{r})\rangle \tag{16}$$

$$= \frac{V^2}{(2\pi)^6}\int_k |\delta(\boldsymbol{k})|^2 e^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{r}}\mathrm{d}\boldsymbol{k}. \tag{17}$$

The power spectrum is defines as:

$$P(\boldsymbol{k}) = \langle |\delta(\boldsymbol{k})|^2\rangle. \tag{18}$$

Then the 2-point correlation function can be expressed by:

$$\xi(\boldsymbol{r}) = \frac{V}{(2\pi)^3}\int_k P(k)e^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{r}}\cdot\mathrm{d}\boldsymbol{k}, \tag{19}$$

from which we see that correlation function is the Fourier transform of the power spectrum.

Instead of 3D, real weak lensing observations are performed on 2D celestial sphere. The 2D density perturbation field can be expressed by:

$$\delta^{2D}(\hat{r}) = \int \mathrm{d}\chi q(\chi)\delta(\chi\hat{r},\chi), \tag{20}$$

where $q(\chi)$ is the lensing kernel which weights the 3D density perturbation at distance χ along the line of sight. We describe it with:

$$q(\chi) = \frac{3H_0^2\Omega_m}{2c^2}g(\chi)\frac{\chi}{a(\chi)}, \tag{21}$$

where $g(\chi)$ is the same as in Eq.(13).

With the Limber's approximation the 2D power spectrum can be written as a line of sight projection of 3D power spectrum:

$$P^{2D}(\boldsymbol{l}) = \int \mathrm{d}\chi \frac{q^2(\chi)}{\chi^2}P^{3D}(|\boldsymbol{k}| = \frac{|\boldsymbol{l}|}{\chi},\chi). \tag{22}$$

From Eq.(12), we see clearly that convergence is a weighted line of sight projection of matter density perturbation. We combine this understanding with the projection of 3D power spectrum in Eq.(22), it's obvious for us to point out that the 2D convergence power spectrum P_κ^{2D} is a weighted line of sight projection of the 3D matter power spectrum $P_{\delta_m}^{3D}$:

$$P_\kappa^{2D}(\boldsymbol{l}) = \frac{9H_0^4\Omega_m^2}{4c^4}\int_0^{\chi_s} \mathrm{d}\chi(\frac{\chi_s-\chi}{\chi_s a(\chi)})^2 P_{\delta_m}^{3D}(|\boldsymbol{k}| = \frac{|\boldsymbol{l}|}{\chi}, \chi). \tag{23}$$

3 Methodology

In this chapter we describe the methodology of this work, including the numerical simulations and the weak lensing maps used in this study, the basic logic of the watershed method, the algorithm for voids merging, calculations of size and ellipticity and two different convergence profile methods.

3.1 Weak lensing maps for algorithm development

3.1.1 Numerical simulation

In this study we use the WL convergence maps generated from N-body simulations taken from Takahashi et al. (2017) which provide full-sky WL convergence maps. The HEALPix resolution of these WL convergence maps is $N_{side} = 4096$ (corresponding to pixel size of 0.86 arcmin), and five source redshifts $z_S = 1.2, 2.4, 3.4, 4.2, 5.3$ (note that in Takahashi simulation the source redshifts are redshifts of fixed single source planes). The N-body simulation has 2048^3 particles, whose mass varies with the box size ranging from 8.2×10^8 to $2.3 \times 10^{12} M_\odot$. A system of nested cubic simulation boxes which had side lengths $L, 2L, 3L, ..., 14L$, with $L = 450h^{-1}Mpc$, was prepared to reproduce mass distribution in the Universe. These boxes are duplicated 8 times and nested around the observer, the nests of smaller boxes are in the center of the nests of larger boxes. The cosmological parameters used in this simulation corresponds to a flat universe with $\Omega_m = 0.279$, $\Omega_\Lambda = 0.721$, $\sigma_8 = 0.820$ and $h = 0.7$.

3.1.2 Galaxy shape noise

The galaxy shape noise (GSN) is the leading source of noise in WL measurements. It's caused by the random shapes and orientations of galaxies, it dominates the observed correlation in galaxy shapes induced by gravitational lensing. In mock data the GSN can be modelled by adding random values drawn from a Gaussian distribution to every pixel of the simulated WL maps (Waerbeke, 2000). The standard deviation of this distribution can be expressed by

$$\sigma_{pix}^2 = \frac{\sigma_{int}^2}{2\theta_{pix} n_{gal}} \tag{24}$$

where σ_{int} is the intrinsic ellipticity dispersion of the source galaxies, θ_{pix} is the width of pixel and n_{gal} is the measured source galaxy number density. In this study we use $\sigma_{int} = 0.4$ and $n_{gal} = 40$ arcmin^{-2}, which match $LSST$ specifications (LSST et al., 2009). However, the noise dominated WL maps can be purified by a Gaussian smoothing process with smoothing length θ_s. If we choose a small value for θ_s, it allows us to maximize the information gained but also leaves significant contamination from GSN. If we choose a large θ_s, the contamination from GSN can be well suppressed but small scale information is lost. This means that the selection of θ_s needs to consider both effects. In this work, in order to explore this fully, we choose 4 smoothing scales, θ_s=1, 2.5, 5, 10 arcmin for the comparisons of the void size and ellipticity statistics with different θ_s in Chapter 4,

the results that include GSN are drawn with dashed lines, the case without GSN are with solid lines.

3.2 Watershed method

The watershed transform is a popular method in image processing, which segments images into distinct regions and objects. This is analogous to a landscape being flooded by a rising level of water. Suppose we have a surface in the shape of mountain terrain, the whole area of this surface can be segmented into different basins, where each basin is associated with a local minima point. The flooding process begins from the lowest local minima, as the water level rises, the surrounded area will be flooded by the water in the expanding basin. When the water level reaches the level of other local minima, these basins also begin the flooding process. After the flooding process is complete, the fully-expanded basins will meet at the ridges which contain the saddle points in the field.
In our work we used Meyer's algorithm to model the flooding process which is outlined as follows:

1. Each local minima pixel is assigned a unique label, these are starting points for the flooding.

2. The neighboring pixels of each marked pixel are inserted into a priority queue with a priority level corresponding to the κ value of the pixel.

3. The pixel with the lowest priority in the queue is extracted, if the neighbors of the extracted pixel that have already been labeled all have the same label (which means are assigned to the same basin), then the pixel is labeled with their label. In contrast, all non-marked neighbors that are not yet in the priority queue are put into the priority queue.

4. Repeat step 3 until the queue is empty. After this step, all pixels should have been assigned to a basin.

Fig.(4) presents the voids identified through Meyer's algorithm, boundaries are along ridges with higher κ values and every local minima pixel belongs to an individual void.

3.3 Merging of voids

After the identification of voids, in order to mitigate the impact of GSN, we compare the average amplitude of the boundary which is shared by two neighbouring voids (κ_{bound}) with the amplitude of their corresponding minima (κ_1 and κ_2). If $\kappa_{bound} - \kappa_1$ or $\kappa_{bound} - \kappa_2$ is smaller than the merge threshold h, we merge two voids into one larger void. Since we use full sky maps, all boundaries are checked in our algorithm to make sure the merge process is finished. We choose $h = \sigma_{GSN}/2$ as the threshold to extenuate the impacts of GSN, we have also tested $h = \sigma_{GSN}$ and $h = 2\sigma_{GSN}$ to find the impact of h on results. Fig.(5) shows the comparison of voids before and after the merging process, the boundary in the green circle disappeared after merging, which indicates the void above merged with the void below.

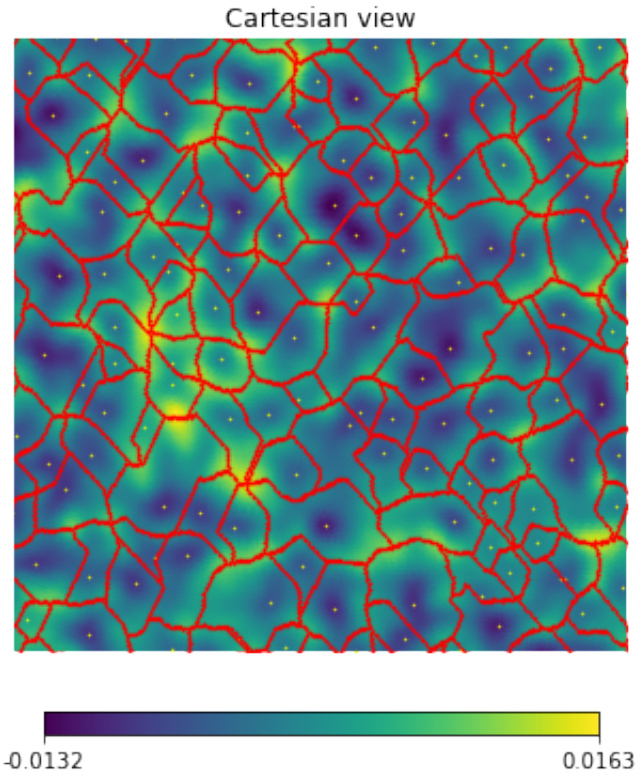
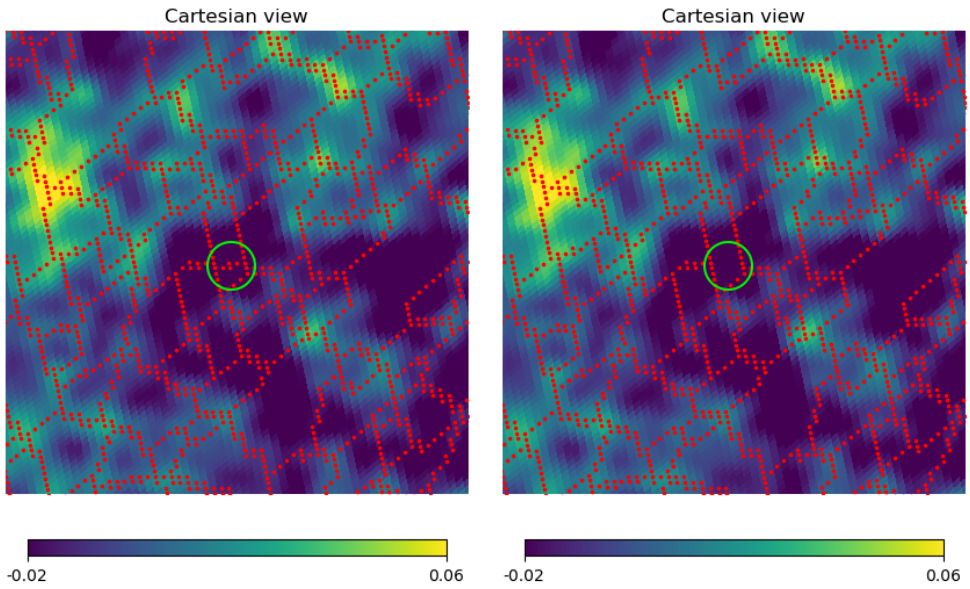


Figure 4: The identified individual voids after the watershed transform, yellow dots are local minima and red lines are the boundaries of voids. The colour of pixels show the value of κ.

3.4 Radius and ellipticity calculation

The outputs of this algorithm are irregular voids which cover the full sky WL convergence map. The latitude and longitude coordinate system of the curved sky can cause problems for the calculation of radius, ellipticity and κ profile. In order to simplify the calculation, we checked the size of the voids and found that the largest void covered 7 deg^2, because the sky coverage of a single void is small enough, we can make a flat sky assumption when we deal with individual voids. The effective radius is calculated with the equation $A = \pi R_V^2$ where A is the area of void.

The ellipticity is more difficult to calculate. First we convert the spherical coordinates of all void pixel to unit vectors in cartesian coordinates from the shell center, we sum these vectors and normalise the result. This gives the centroid of void. The reason for converting the coordinates is that in spherical coordinates longitudinal neighbouring pixels with higher latitude have a larger difference than longitudinal neighbouring pixels which have lower latitude, this effect is very obvious for the voids near the poles. If we just calculate the mean of the void pixel's latitude and longitude coordinates, we will get the wrong coordinates for the centroid. After the centroid is calculated, we create a plane which is perpendicular to the vector from the shell center to the centroid, all void pixel are then projected to this plane. Because of the flat sky assumption we can still assume that the



(a) Voids before merging process (b) Voids after merging process

Figure 5: The effect of merging

projected pixel on the 2D plane are homogeneously distributed. Fig.(6) presents the effect of this projection process, the selected void is shown in the left figure with yellow dots and the boundary with red dots. The figure on the right side is the projected void in the flat 2D plane, the orientation has changed because we use orthogonal basis vectors for projection.

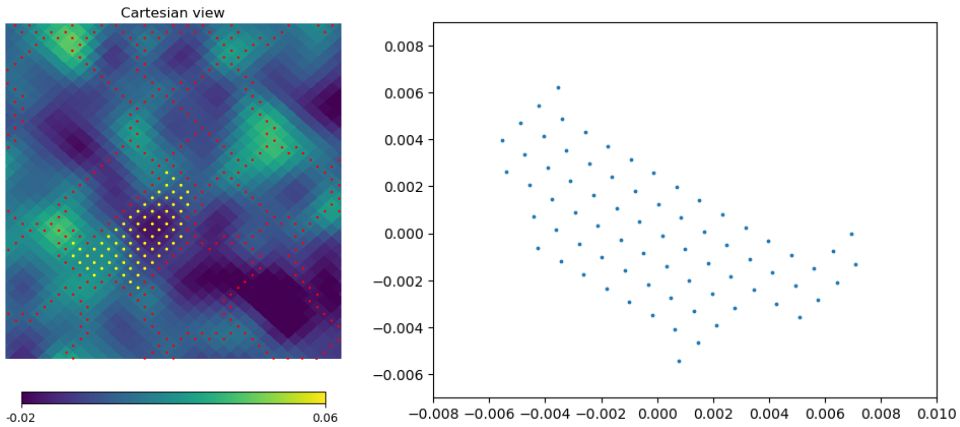
We used the definition of ellipticity ϵ in the work of Park and Lee (2007), first we create a pseudo inertia momentum matrix $I_{i,j}$ for the selected void. It can be expressed as

$$I_{i,j} = \frac{1}{N_{pix}} \sum x_i \cdot x_j \tag{25}$$

where N_{pix} is the number of pixels in the selected void, i is the dimension of the coordinates, and x_i is the position of a pixel in the i-th dimension. Because what we are interested in is the geometry of voids, the computation of $I_{i,j}$ is not weighted with the κ value of every pixel.
We can find two eigenvalues λ_1, λ_2 of the matrix $I_{i,j}$, which allows us to write the ellipticity as

$$\epsilon = 1 - (\frac{\lambda_1}{\lambda_2})^{0.5} \tag{26}$$

where $\lambda_1 \leq \lambda_2$.



(a) Void on curved convergence map

(b) Projected void on 2D plane

Figure 6: The effect of projection

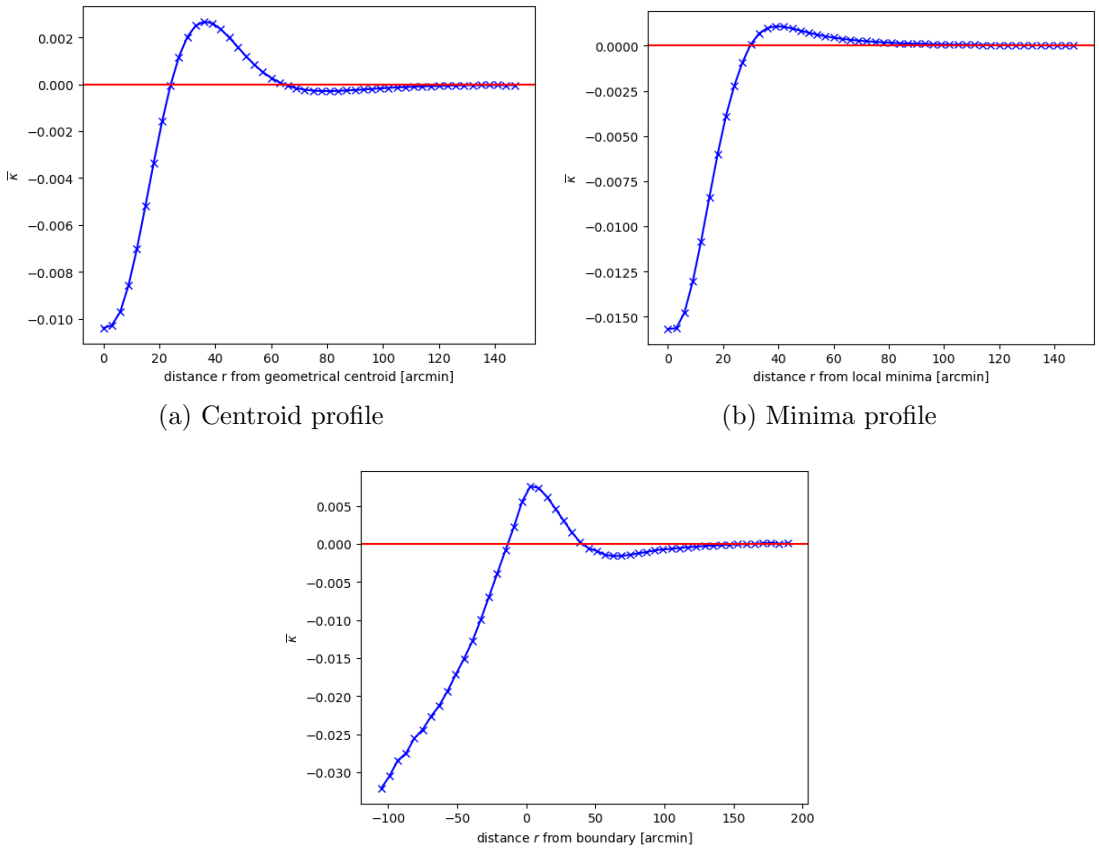
3.5 The convergence profile of voids

In this part we introduce two different methods for computing void convergence profiles. The first method is based on the distance to center point inside void, the second method calculates with the distance to boundary.

3.5.1 Radial profile

In the Radial profile method the starting point is the geometrical center (centroid) of every void, the profile is computed as a function of the radial distance and the rescaled distance (divided by the void radius) from the centroid. There are also other potential choices for the center, for example the local minima and the κ weighted center. The resulting profile can be slightly affected by the choice of center.

Because of the same coordinate system problem we described in the last subsection, the void is first projected to a 2D plane, the void centroid is given by $\boldsymbol{x}_c = \sum \boldsymbol{x}_i/N$, where $\boldsymbol{x}_i$ are the coordinates of i-th pixel in the void and N is the number of pixels inside the void. After we calculate the centroid coordinates in the 2D plane we find the nearest pixel, this pixel is the centroid for the profile computation. In order to compare the difference caused by different definitions of center we also used local minima for the computation, the void which is merged from two or more voids will choose the minima with lowest κ as its new local minima point.



(a) Centroid profile

(b) Minima profile

(c) Boundary profile

Figure 7: Comparison of three different profiles.

Fig.(7) (a), (b) shows the results of two different radial profile methods with the same map and parameters. The κ value for the starting of the point of centroid profile is larger than the minima profile, because most centroids are not located at the same point as the local minima. The maximal κ of the centroid profile is higher than the minima profile, the reason is that with centroids as starting points, it's more likely to collect boundary pixel in the same annulus (see Fig.(8)).

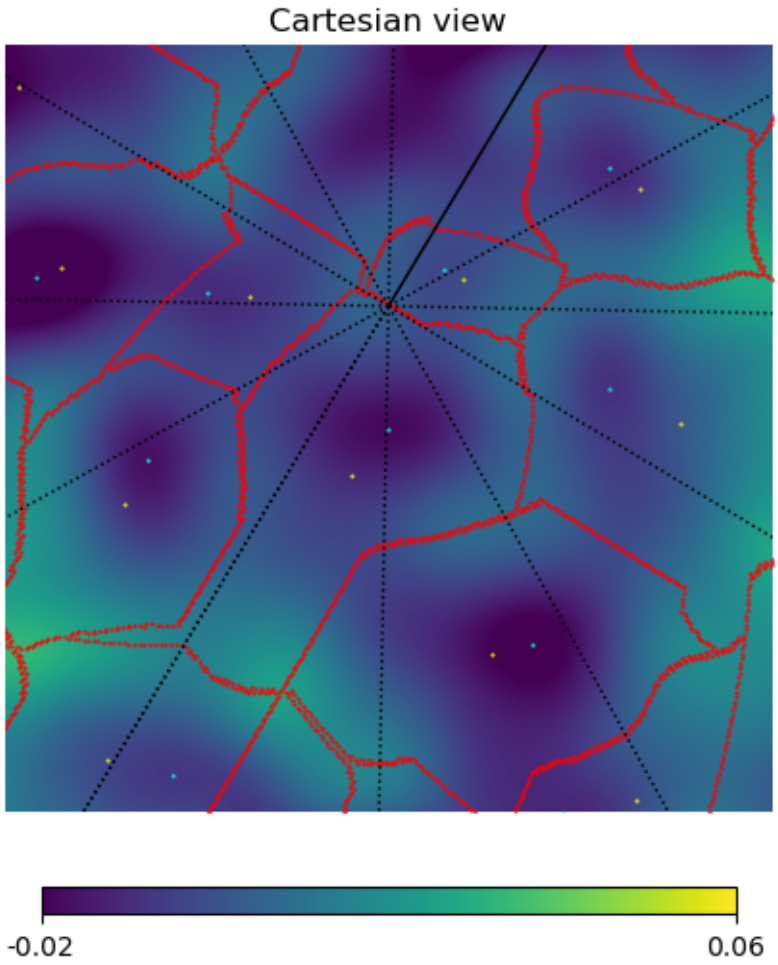


Figure 8: Blue dots are local minima and yellow dots are centroids.

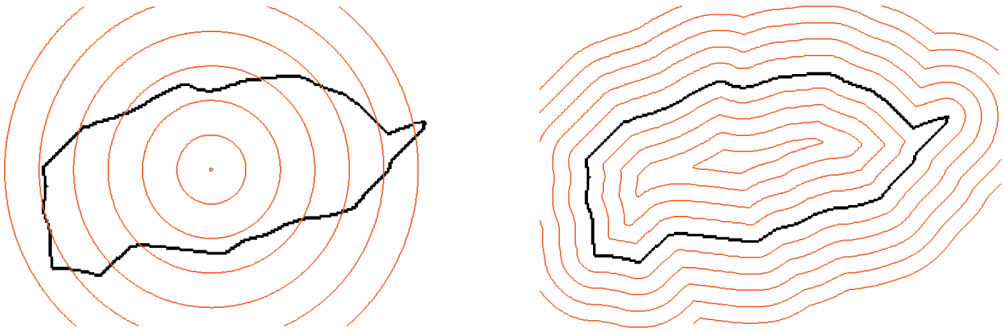
3.5.2 Boundary profile

Cautun et al. (2016) presented a new void profile computation method. Instead of the centroid, the boundary is used as the starting point for the convergence profiles. The profile is computed by the shortest distance from the pixel to the boundary of selected void. In order to differentiate whether the profile is inward or outward from the void boundary, we give a minus sign to the distance of the pixel inside the selected void and a positive sign to the distance of the pixel outside the void. This process can be expressed with the following equation:

$$D = \begin{cases} + \, min_i \, [\, |\boldsymbol{x} - \boldsymbol{y}_i| \,] & \text{for } \boldsymbol{x} \text{ outside of void} \\ - \, min_i \, [\, |\boldsymbol{x} - \boldsymbol{y}_i| \,] & \text{for } \boldsymbol{x} \text{ inside of void} \end{cases} \tag{27}$$

where D corresponds to the distance from the pixel to the boundary, $\boldsymbol{x}$ denotes the Cartesian coordinates of the pixel and $\boldsymbol{y}_i$ denotes to the coordinates of the boundary pixel.

The advantage of boundary void profiles is the independence of void shape. In the radial profile method, as the radial distance from the center increases, the annuli intersect



(a) Radial profile (b) Boundary profile

Figure 9: Illustration of two different profile measuring methods.(Cautun et al., 2016)

the void boundary, which results in the void boundary appearing in multiple annuli (see Fig.(9)). The extent to which this occurs the shape of the void, if the void is circular, then this impact can be neglected, in contrast, if the void is long and narrow, the distribution of the background density dominates the profile.

This is shown in Fig.(7) (c) where the boundary profile has the highest convergence contrast in all three profiles which proves that the boundary profile has "cleaner" features than the radial profile.

3.6 Application with non full-sky DES data

This void finding algorithm is also compatible with real survey data. We applied this code to real convergence maps, which were reconstructed from DES Year 3 data sets by Jeffrey et al. (2021). The maps were reconstructed with Kaiser-Squires (KS) method (Jeffrey et al., 2021), the convergence κ was derived from the Fourier-transform of the WL shear γ of the pixel nearby. There are four tomographic bins in the mass map (see Fig.(10)).

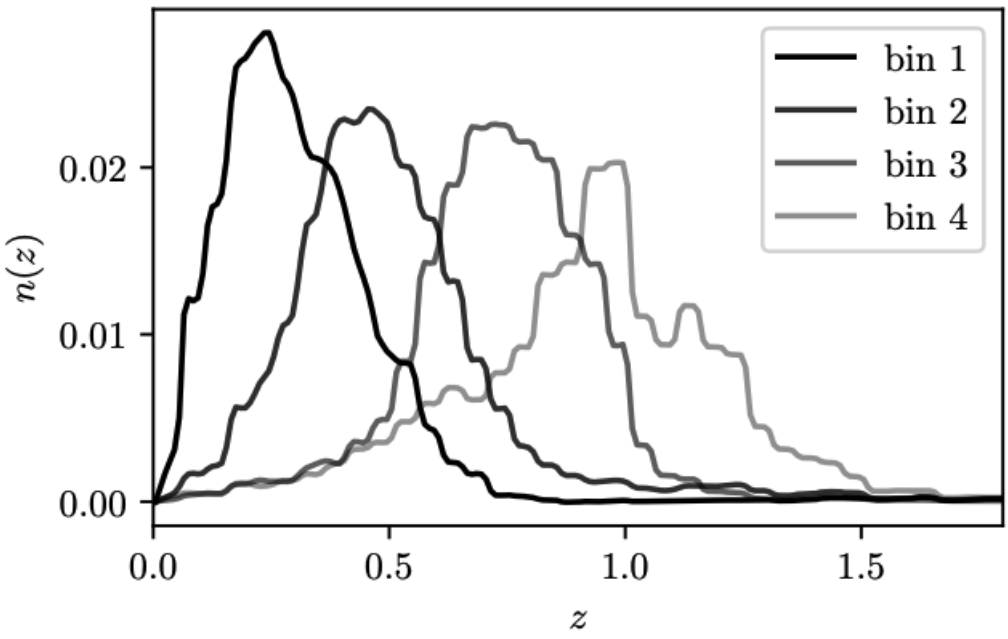
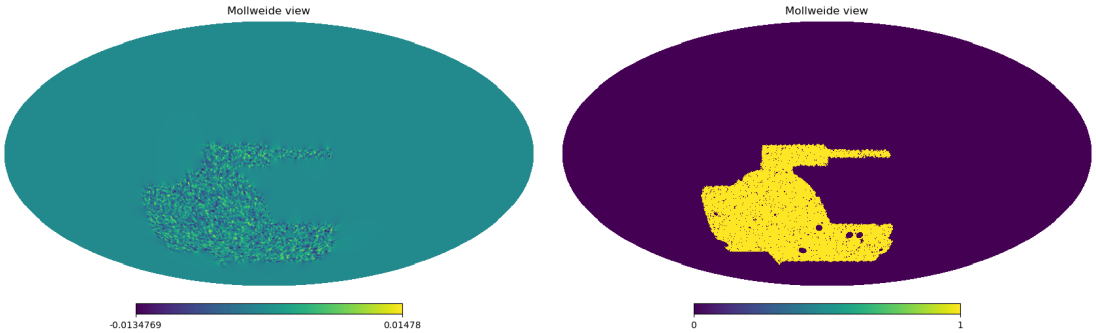


Figure 10: Redshift distributions as estimated in data for the four DES Y3 tomographic bins (Myles et al., 2021).

Fig.(11) (a) shows the WL mass map from the DES Y3 data, which covers 4143 deg^2,



(a) (b)

Figure 11: The convergence map and the mask.

Fig.(11) (b) shows the mask for the DES Y3 data, the pixels with observed information

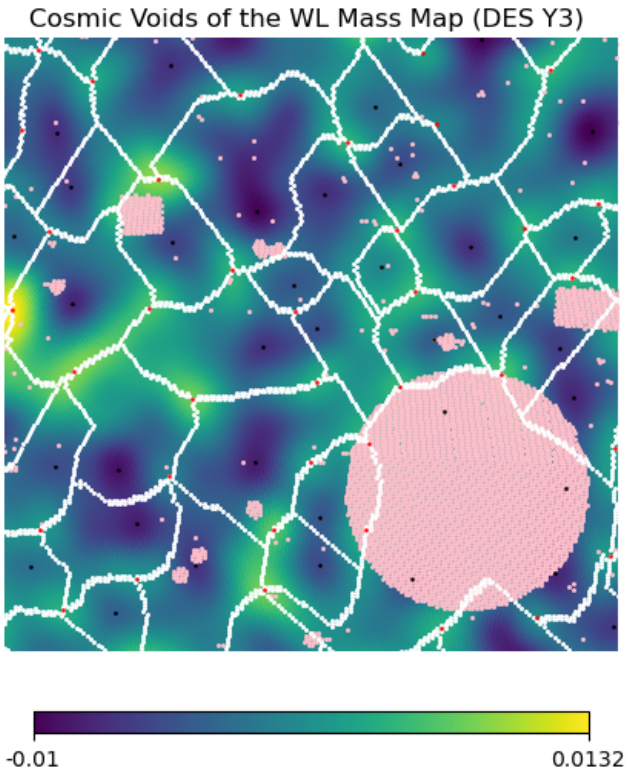
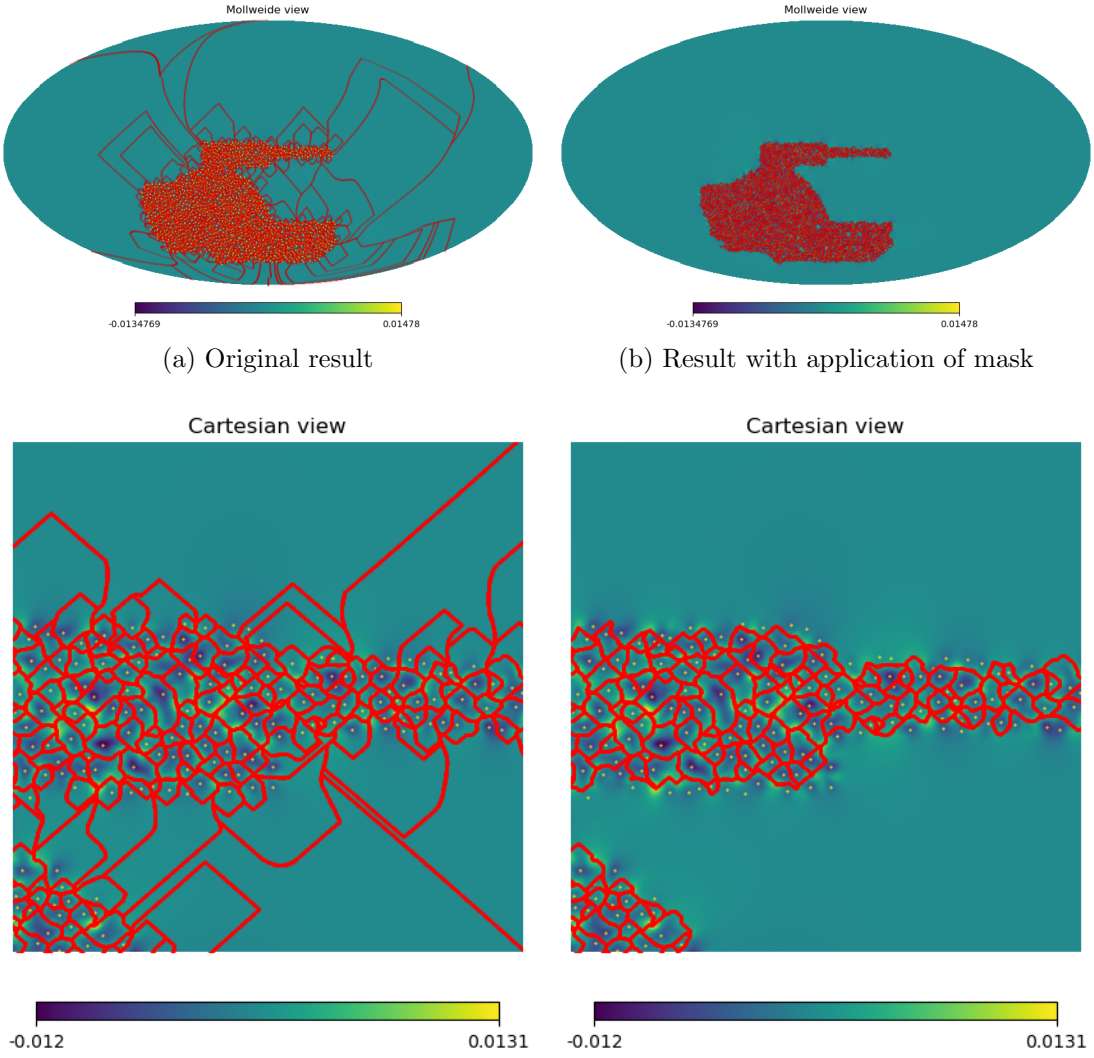


Figure 12: Zoom in of result covered by mask

are assigned a value of 1 and empty pixel are assigned a value of 0 in the mask file.

In Fig.(12) the red dots are local maxima (all of them are locate on ridges or knots) and the blue dots are local minima, pink dots correspond to masked pixels. As we can see, most of the pixels of the big voids on the lower right are masked pixels, which means these pixels have fewer information. The high proportion of masked pixels in the DES mass map makes the result of original algorithm unreliable. Fig.(13) (a) and (c) present the original result, because every pixel has a value, the regions without usable information are also included in some voids nearby.

In order to solve this problem, we counted the proportion of masked pixels in each void, then we set a cut-off threshold, if the proportion is larger than the threshold, this void will be deleted. Fig.(13) (b) and (d) show the result with the application of 10% cut-off threshold, the "fake" WL voids are removed, the rest of the voids are all in reasonable regions. We also tested two other thresholds 30% and 50%, found that there are few difference among the results of three thresholds. The results are presented in Chapter 5.



(a) Original result

(b) Result with application of mask

(c) Zoom in of original result

(d) Zoom in of result with application of mask

Figure 13: Identified basins without and with mask.

3.7 Application with CosmoGridV1 data

CosmoGridV1 comprises an extensive collection of lightcone N-body simulations aimed at map-level cosmological inference utilizing large scale structure probes. Its purpose is to facilitate cosmological parameter measurements through Stage-III photometric surveys, incorporating non-Gaussian statistics and machine learning techniques. CosmoGridV1 spans the space of the wCDM model by varying Ω_m, σ_8, w_0, n_s, Ω_b and H_0, it is covered by 2500 grid points on a Sobol sequence (see Fig.(14)). There are 7 simulations for each grid cosmology and 200 independent simulations for the fiducial cosmology. The baryon feedback effects are added with shell-based baryon correction model on the map level (Kacprzak et al., 2023). The weak lensing full-sky map can be created with these shells. The projected map has $nside = 512$ and each node has four tomographic bins, which correspond to the redshift distribution of the weak lensing sources (see Fig.(15)).

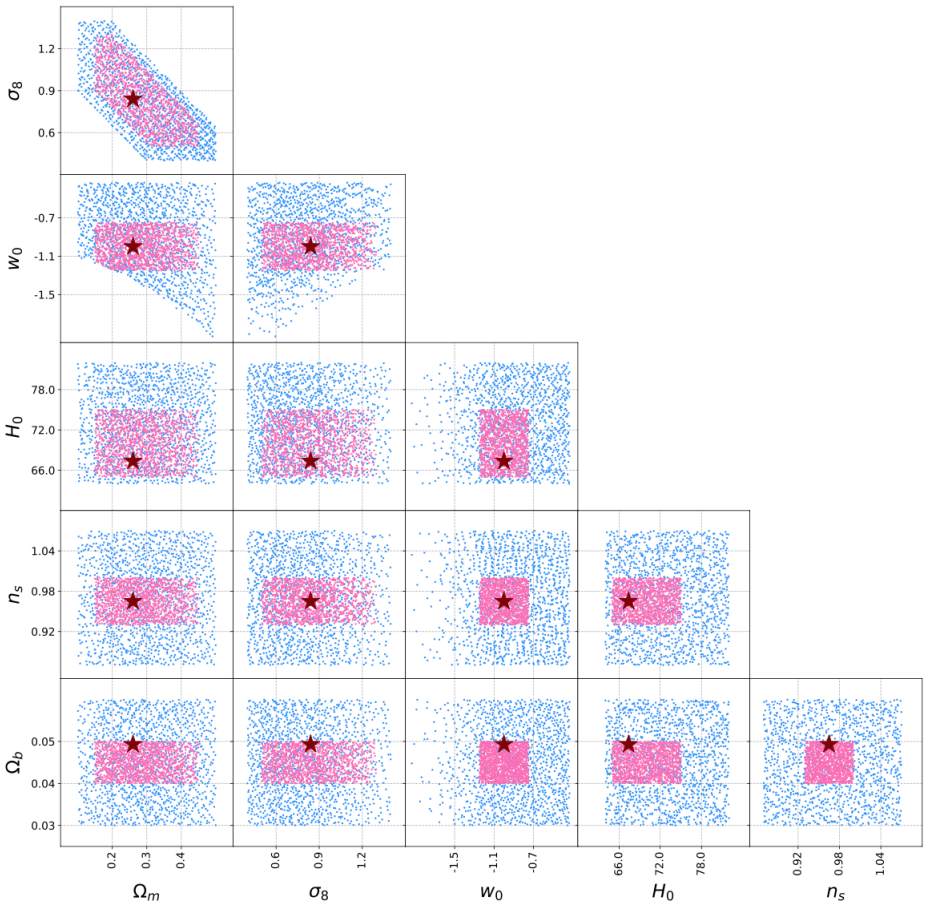


Figure 14: 2500 five dimensional cosmological parameters used in CosmoGridV1. The blue and pink parameters belong to the "wide" and "narrow" grids, respectively. (Kacprzak et al., 2023)

We apply the void finder code to all WL convergence maps for 2500 grid cosmologies (70000 individual maps) and 200 maps for the fiducial cosmology, one of the results is shown in Fig.(16). Due to the large total computing time, we use the smoothing scale

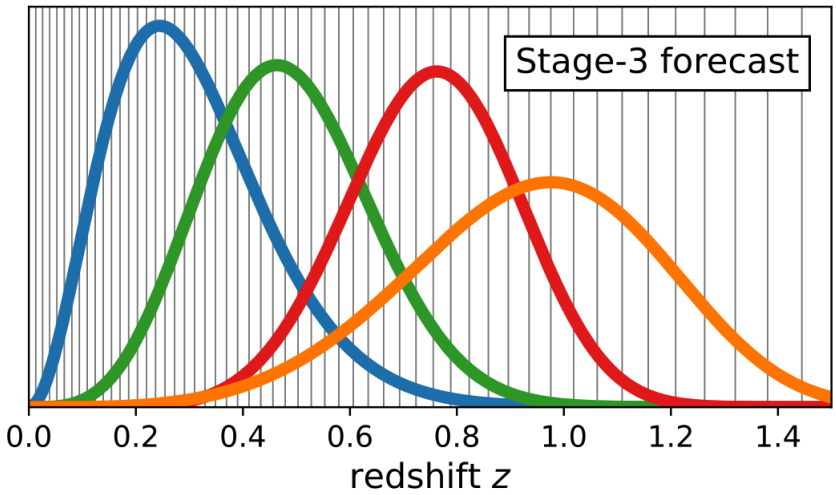


Figure 15: The tomographic redshift distribution of weak lensing sources. (Fischbacher et al., 2023)

$\theta_s = 10$ arcmin and a merge threshold of 0.5 GSN for the void finding. We also only calculate the size and the ellipticity distribution for the identified voids to save on computation time.

In order to make predictions of the WL void statistics at arbitrary points in the 6D parameter space shown in Fig.(14), we use a Gaussian process regression (GPR) emulator from scikit-learn (Pedregosa et al., 2018) to interpolate the void statistics between different cosmologies. The accuracy of the GPR emulator is limited by the availability of training data, because it needs the training data to sample the parameter space sufficiently. The average void statistics and their standard errors at each cosmology are used as the training data for the emulator. There are 4 emulators for voids size statistics and 4 emulators for voids ellipticity statistics, each emulator corresponds to a single tomographic bin.

After the training of the emulator, we use Monte Carlo Markov Chain (MCMC) to estimate the posteriors of the parameters for the entire parameter space and produce likelihood contours. The MCMC analysis is conducted with the emcee python package (Foreman-Mackey et al., 2013).

We use the Bayesian formalism, the likelihood $P(p|d)$ of the set of cosmological parameters $\boldsymbol{p} = [\Omega_m, \sigma_8, w_0, n_s, \Omega_b, H_0]$ given a data set $\boldsymbol{d}$, is given by

$$P(\boldsymbol{p}|\boldsymbol{d}) = \frac{P(\boldsymbol{p})P(\boldsymbol{d}|\boldsymbol{p})}{P(\boldsymbol{d})} \tag{28}$$

where $P(\boldsymbol{p})$ is the prior, $P(\boldsymbol{d}|\boldsymbol{p})$ is the likelihood of the data conditional on the parameters and $P(\boldsymbol{d})$ is the normalisation. In this work we use flat priors with lower and upper limits for $\Omega_m : [0.10, 0.50]$, $\sigma_8 : [0.40, 1.40]$, $w_0 : [-2.00, -0.33]$, $n_s : [0.87, 1.07]$, $\Omega_b : [0.03, 0.06]$, $H_0 : [64.0, 82.0]$. These priors match the parameter space described in Fig.(14).

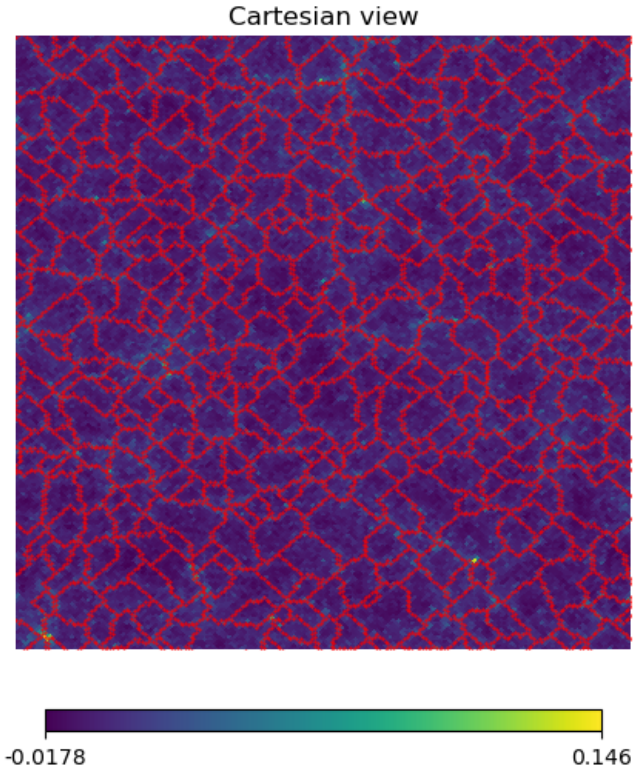


Figure 16: The identified voids of CosmoGridV1 WL map.

With the equation

$$log(P(\boldsymbol{d}|\boldsymbol{p})) = -\frac{1}{2}[\boldsymbol{d} - \mu(\boldsymbol{p})]C^{-1}[\boldsymbol{d} - \mu(\boldsymbol{p})] \tag{29}$$

we can express the log likelihood, where $\mu(\boldsymbol{p})$ is the prediction generated by the emulator for a set of parameters $\boldsymbol{p}$, and C^{-1} is the inverse covariance matrix. We use the emulator's prediction of a statistic at the fiducial cosmology as the data $\boldsymbol{d}$. The likelihood returns a 6D probability distribution that indicates how well different regions of the parameter space describe the input data $\boldsymbol{d}$ (Davies et al., 2021).

In practice we use three different statistics as the training data, the first one is the void size function data, the second one is the ellipticity distribution, the third one is the combination of the two. We combine all tomographic bins for the training data, in order to improve the constraint power. The constraint forecasts will be shown and discussed in Chapter 5.

4 Weak Lensing Voids Statistics

In this section we present and discuss the statistics measured from the watershed void catalogue, the abundance in Section 4.1, the ellipticity in Section 4.2, and the convergence profile in Section 4.3. Fig.(17) and Fig.(18) are 2D histograms of size and ellipticity distributions of 2D WL voids. Tab.(1) shows the statistical values for the comparison of different redshift, Tab.(2) provides a quantitative comparison of different smoothing scales.

Table 1: The range of the effective radius (R_V^{max} and R_V^{min}), the mean effective radius ($\overline{R_V}$), the mean ellipticity ($\overline{\epsilon}$), the mean maximal and minima convergence ($\overline{\kappa_{max}}$ and $\overline{\kappa_{min}}$), and the mean number of identified voids ($\overline{n_V}$).

z	R_V^{max}	R_V^{min}	$\overline{R_V}$	$\overline{\epsilon}$	$\overline{\kappa_{max}}$	$\overline{\kappa_{min}}$	$\overline{n_V}$
5.3	1.346	0.046	0.477	0.360	0.016	-0.016	44265
4.2	1.379	0.056	0.478	0.360	0.015	-0.014	44057
3.4	1.387	0.056	0.479	0.361	0.013	-0.013	43895
2.4	1.442	0.046	0.480	0.362	0.011	-0.010	43650
1.2	1.450	0.046	0.482	0.365	0.007	-0.006	43250

Table 2: The range of the effective radius (R_V^{max} and R_V^{min}), the mean effective radius ($\overline{R_V}$), the mean ellipticity ($\overline{\epsilon}$), the mean maximal and minima convergence ($\overline{\kappa_{max}}$ and $\overline{\kappa_{min}}$), and the mean number of identified voids ($\overline{n_V}$).

Smoothing scale	R_V^{max}	R_V^{min}	$\overline{R_V}$	$\overline{\epsilon}$	$\overline{\kappa_{max}}$	$\overline{\kappa_{min}}$	$\overline{n_V}$
10	1.379	0.056	0.478	0.360	0.015	-0.014	44057
5	0.852	0.032	0.240	0.383	0.019	-0.018	152274
2.5	0.461	0.032	0.128	0.440	0.025	-0.022	430844
1	0.342	0.032	0.085	0.509	0.033	-0.029	801602

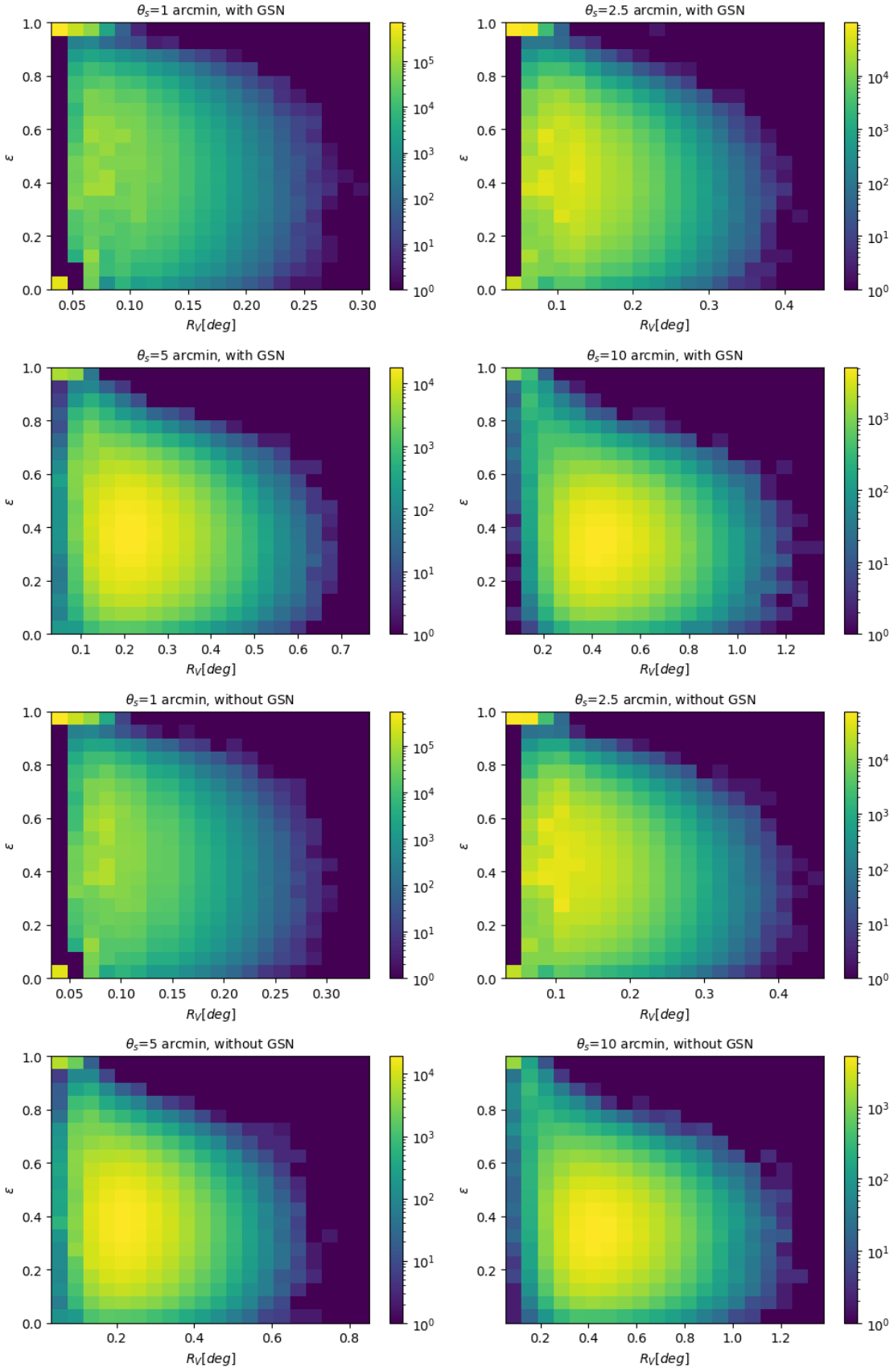


Figure 17: The 2D histogram of void size and ellipticity distribution with different smoothing scale. The redshift is 4.2 and the merge threshold is 0.5 GSN.

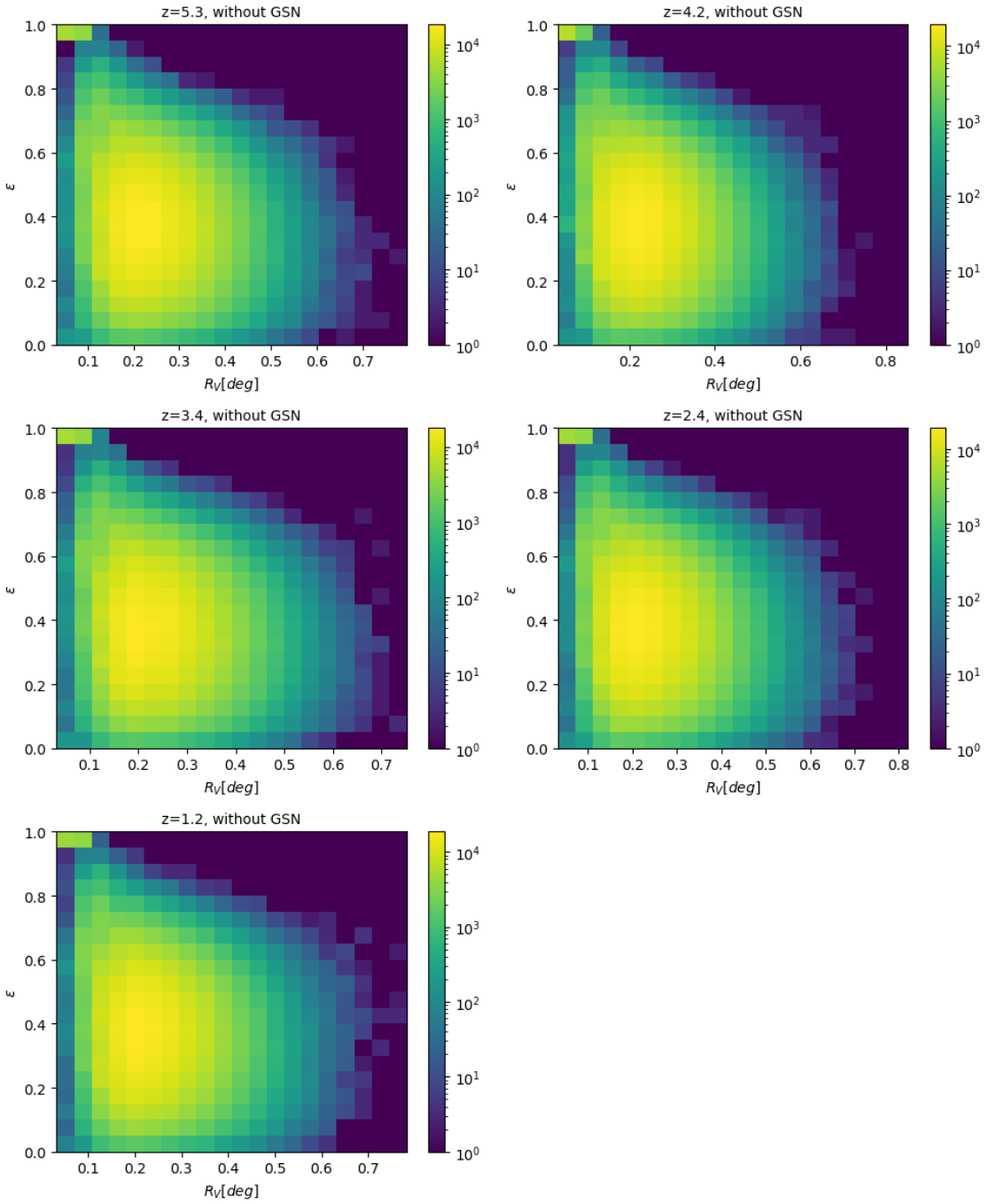


Figure 18: The 2D histogram of void size and ellipticity distribution with different redshift. The smoothing scale is 5 arcmin and the merge threshold is 0.5 GSN.

4.1 Void abundance

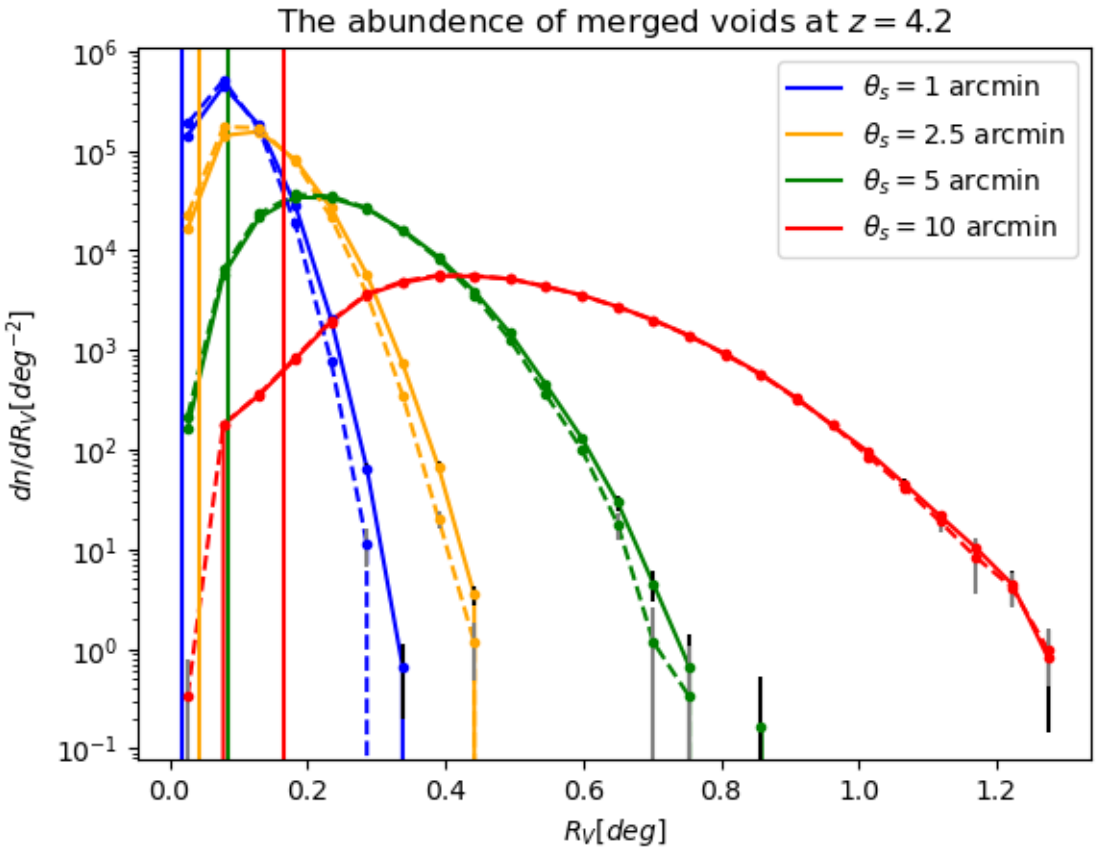


Figure 19: The differential void size abundance as a function of void radius R_V with different smoothing scale. The solid lines correspond to the statistics of maps without noise and the dashed lines correspond to the statistics of maps with noise. The vertical solid lines correspond to 4 different smoothing scales.

The differential void size abundance as a function of void radius R_V with different smoothing scales (1, 2.5, 5 and 10 arcmin) are shown in Fig.(19). The four different colours correspond to four smoothing scales θ_s=1, 2.5, 5 and 10 arcmin. As the smoothing scale increases, the shape of the curve flattens out, which means the total number of voids decreases and the void sizes increases, in Tab.(2) this trend is very obvious. We can also find that the case with GSN has more small-radius voids and less large-radius voids, in the results of smoothing scale 1, 2.5 and 5 arcmin there are moderate disagreements between the GSN maps and the no GSN maps, but in the distributions of smoothing scale 10 arcmin we can barely find it. The reason is that the GSN increases the fluctuations in convergence map, more fluctuations means more watershed voids are identified, meanwhile higher smoothing scales erase more fluctuations in the convergence map which leads to fewer and larger voids.

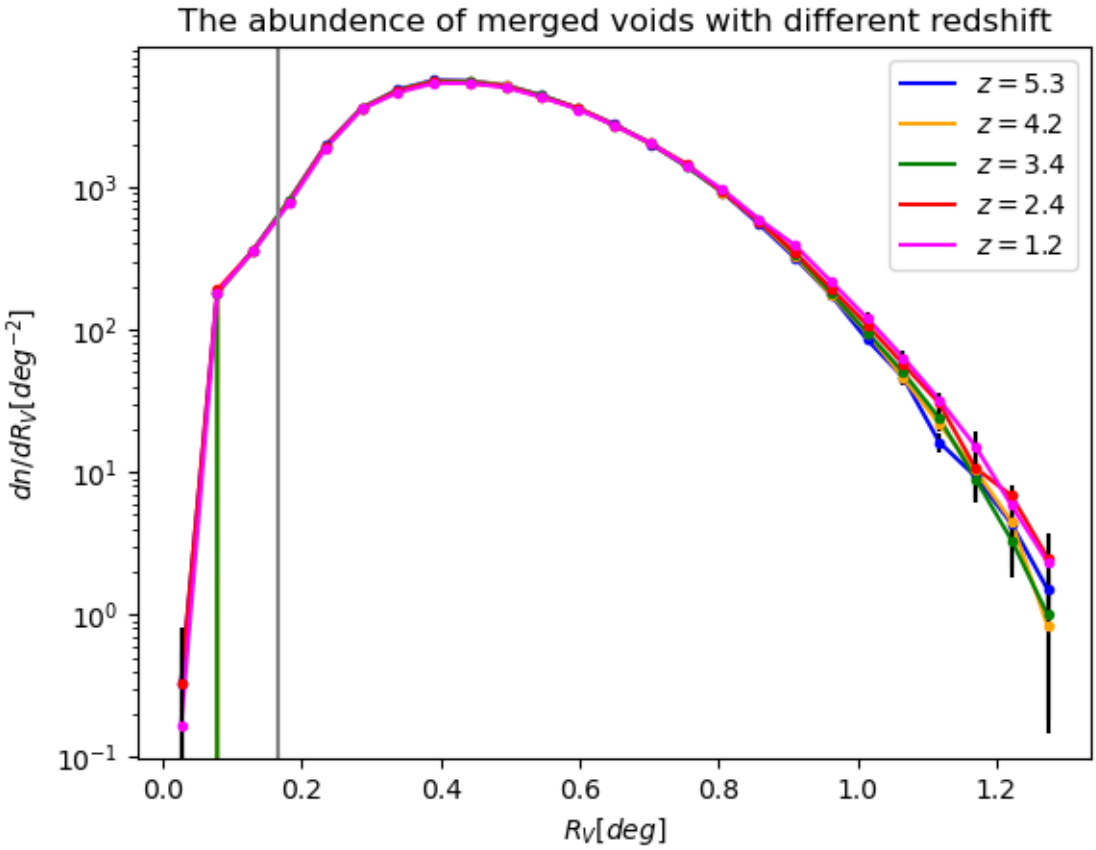


Figure 20: The differential void size abundance as a function of void radius R_V with different red shift. The vertical solid line corresponds to smoothing scale $\theta_s = 10$ arcmin.

Fig.(20) shows the differential void size abundance of the maps with different redshift, the most abundant voids of all redshifts are the voids with around R_V=0.4 deg. In the right part of curves we can find that the lower redshifts have more large voids than the higher redshift. in Tab.(1) we can confirm this trend, as the redshift z rises, R_V^{max} and $\overline{R_V}$ rise, meanwhile $\overline{n_V}$ decreases. Note that the convergence κ is the integral along the line-of-sight, a lower redshift means less mass projection, which leads to smaller κ perturbations (i.e. less structure) on the map, thus the voids become larger in low redshift, $\overline{\kappa_{max}}$ and $\overline{\kappa_{min}}$ in Tab.(1) support this explanation. Previous researches also show that the 3D voids are larger at low redshift as they have had more time to expand. This means that their imprint on the convergence map will correspond to larger angular scales (Jennings et al., 2013, Contarini et al., 2022).

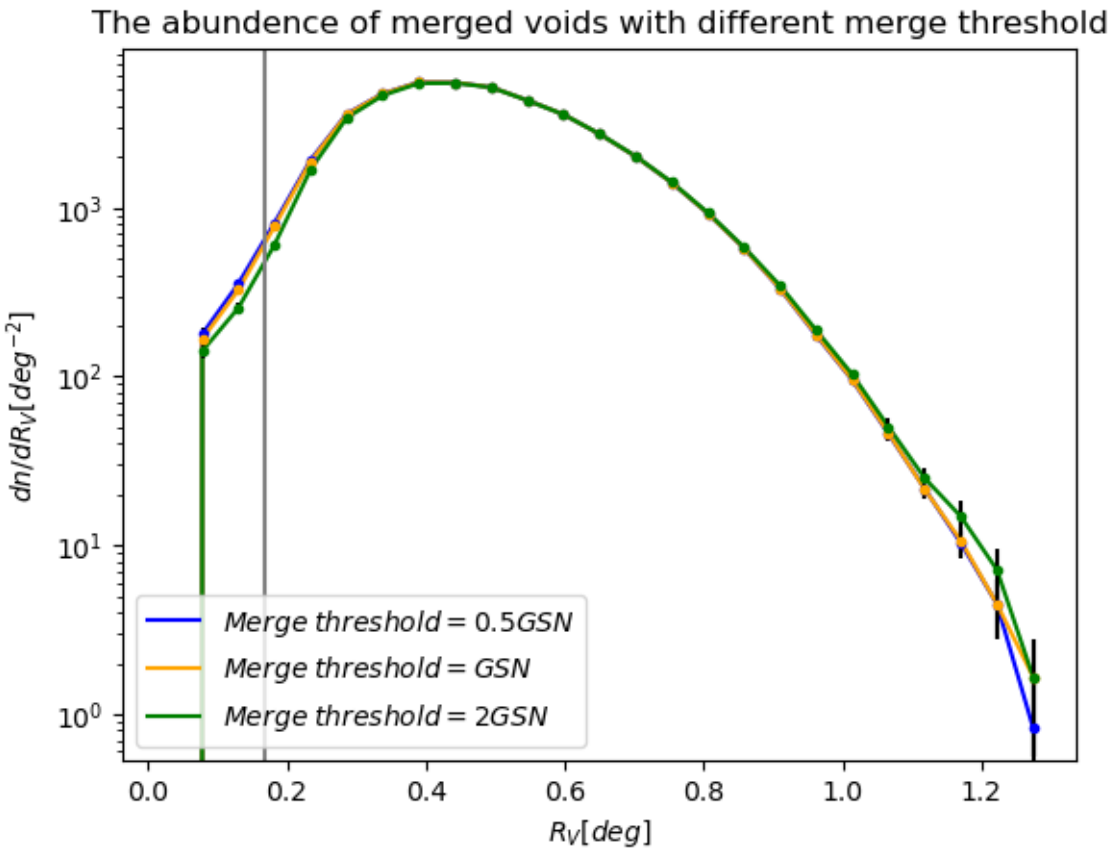
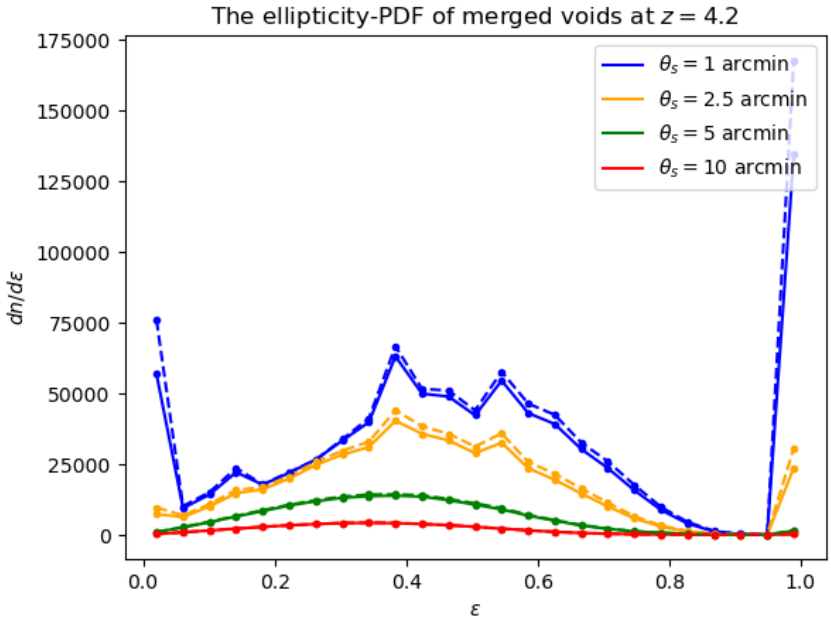


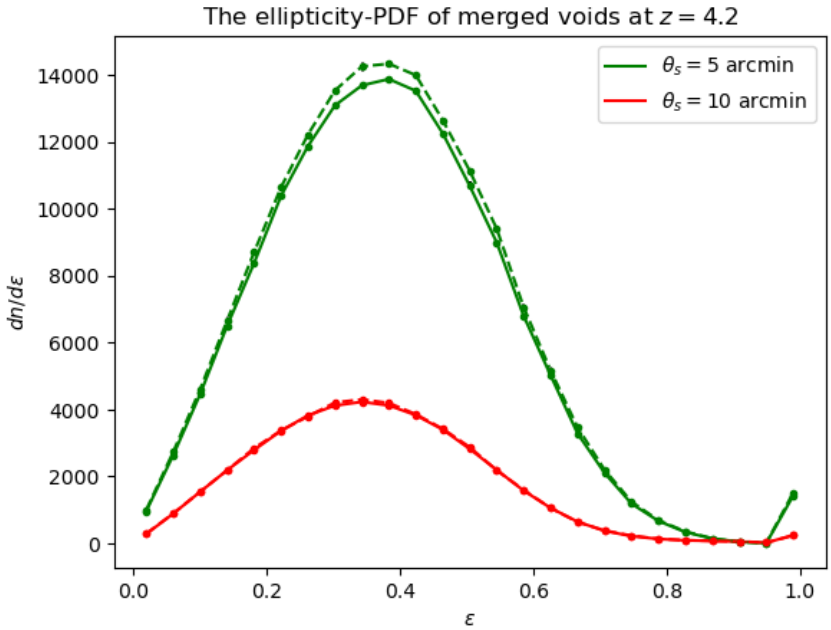
Figure 21: The differential void size abundance as a function of void radius R_V with different merge threshold. The vertical solid line corresponds to smoothing scale $\theta_s = 10$ arcmin.

The statistics for WVF voids (z=4.2) merged with different merge thresholds are shown in Fig.(21). The most left part of green curve is lower and the most right part is higher than the other two curves. It indicates that the higher merge threshold decreases the abundance of small voids and increases the abundance of large voids, which can be explained by the methodology of merge. A higher merge threshold means that more boundaries can be identified as the noise caused by GSN and it leads to more merging events. We can also find that the difference is not significant in this figure, the reason might be the large smoothing scale we used, which leads to loss of information.

4.2 Void ellipticity



(a) All four smoothing scales



(b) Two reliable smoothing scales

Figure 22: The void ellipticity-PDF as a function of void ellipticity ϵ with different smoothing scale. The dashed lines correspond to the statistics of maps with noise.

Fig.(22) (a) plots the void ellipticity distributions of redshift 4.2 with four different smoothing scales. The statistics of 1 and 2.5 arcmin have uneven shapes, both have high amounts of high-elliptical voids (statistics of 1 arcmin also has many low-elliptical

voids), this indicates 1 and 2.5 arcmin are too small to suppress the Noise on the maps. Thus we select more reliable smoothing scales for later analysis. It's also worth noting that the ellipticity distribution moves towards low ellipticity as the smoothing scale increases (see $\bar{\epsilon}$ in Tab.(2)). This indicates that the larger a void is, the more spherical it is. In Fig.(22) (b) shows ellipticity distributions for 5 and 10 arcmin, the curves are more like a Gaussian distribution, the dashed curves show that GSN doesn't change the ellipticity distribution significantly.

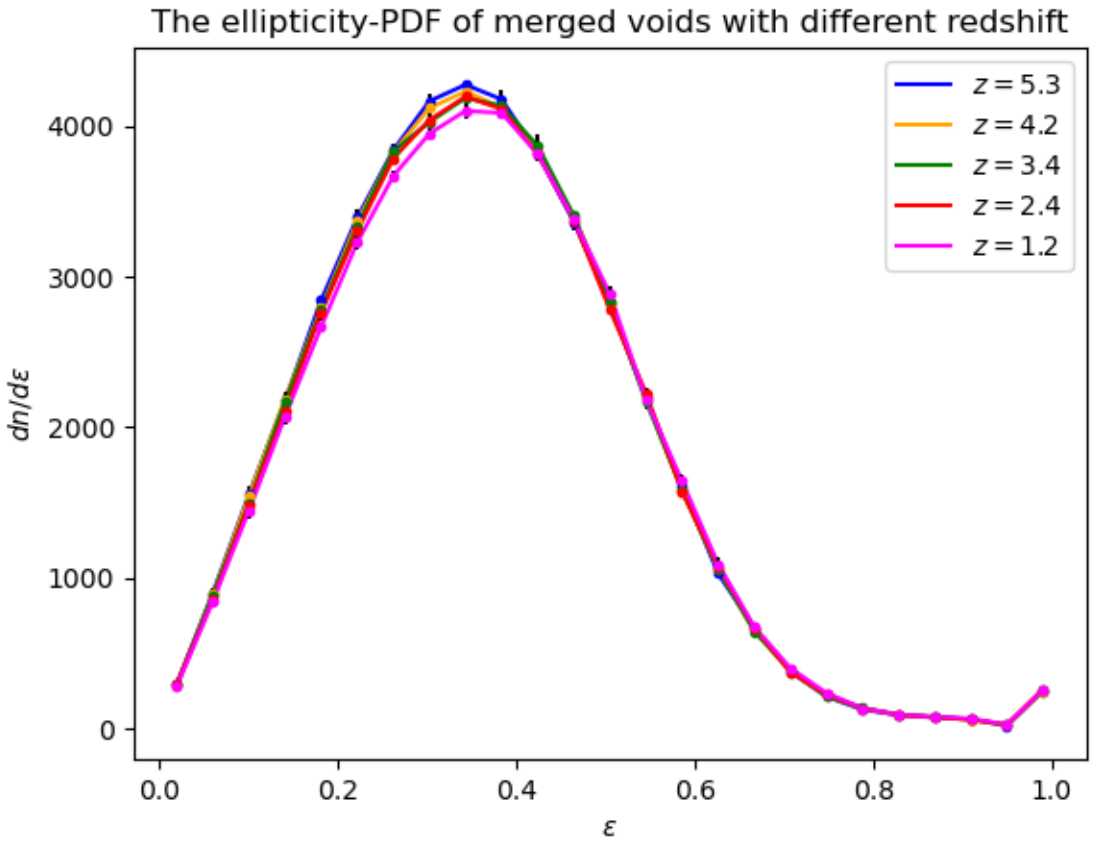


Figure 23: The void ellipticity-PDF as a function of void ellipticity ϵ with different red shift.

In Fig.(23) we can see the ellipticity distribution of merged voids with different redshifts. The rough shape doesn't change much with redshift, the most abundant ellipticity is around 0.35. We can observe a small trend, as the redshift decreases the value of most abundant ellipticity increases. The same trend also appeared in previous work of 3D voids (Bos et al., 2012). The mean ellipticity of the voids increases with time, with voids being less elongated towards higher redshifts. The theoretical formalism of this trend is presented in the work of Lee and Park (2009).

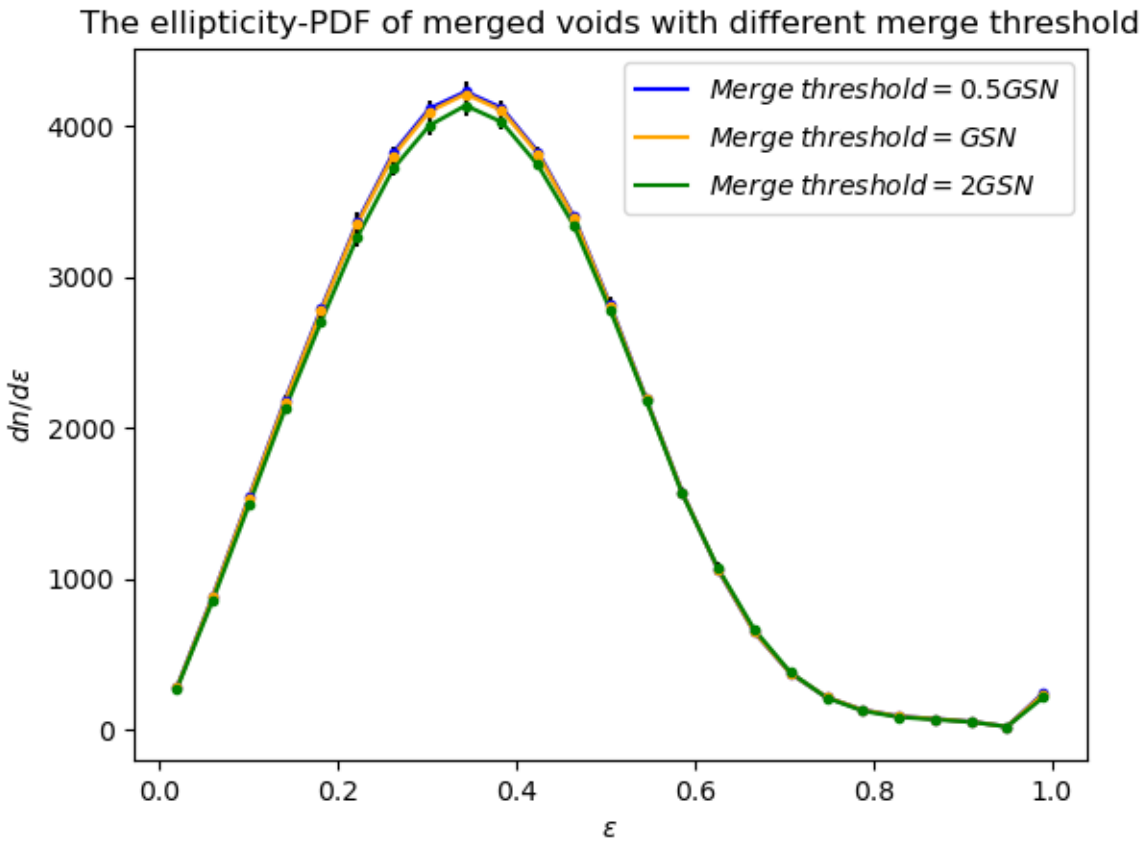


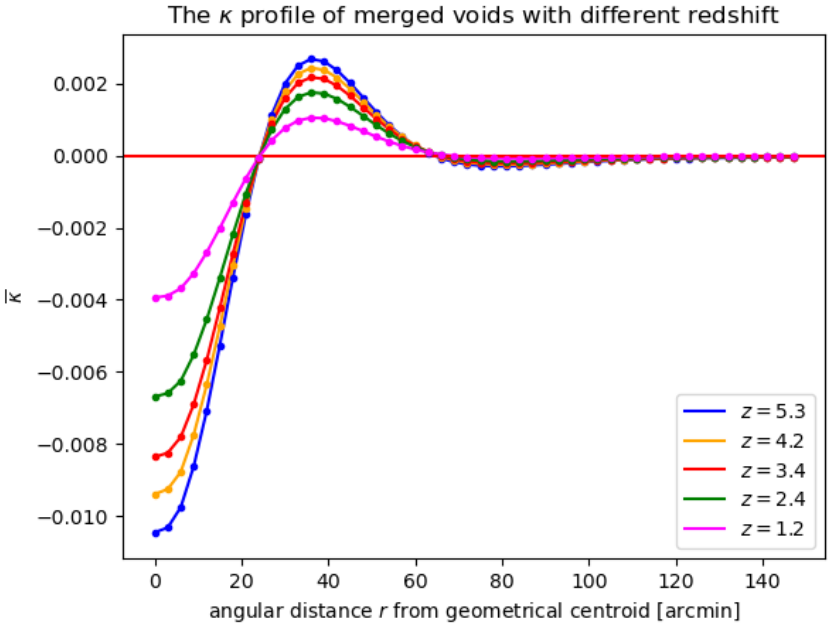
Figure 24: The void ellipticity-PDF as a function of void ellipticity ϵ with different merge threshold.

The ellipticity distributions of different merge thresholds are in Fig.(24). All three peaks locate at $\epsilon \sim 0.35$ and the shape of the distribution doesn't change with merge threshold which means the merging process of weak lensing voids has nearly no influence on the ellipticity distribution.

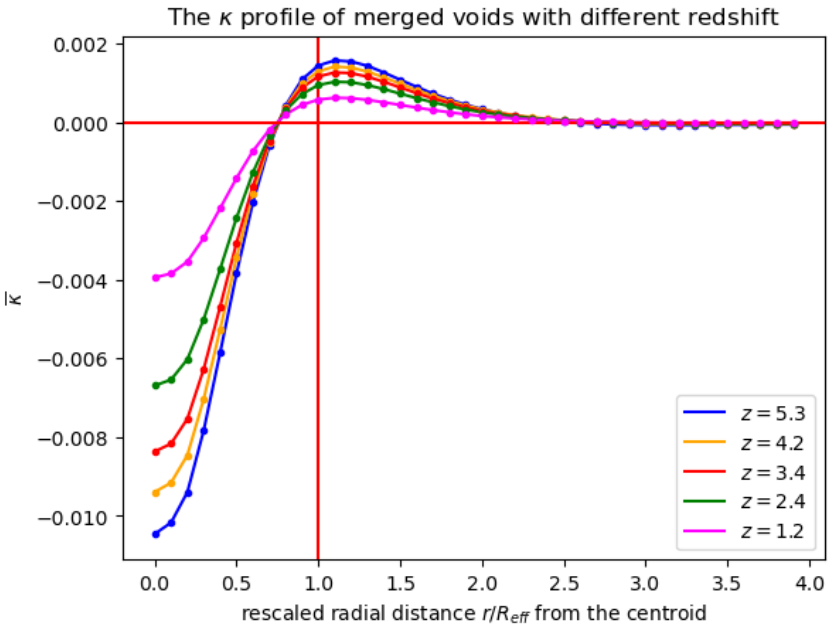
4.3 Void convergence profile

We applied 2 different profile analysis methods to weak lensing voids for redshift $z = 4.2$ with GSN.

4.3.1 Centroid profiles



(a)



(b)

Figure 25: The convergence profile from void's centroid with different red shift.

In Fig.(25) (a) we present the centroid method convergence profiles of voids from five different redshifts. They have a smooth shape with negative convergence values at $r = 0$, κ increases outwards and crosses $\kappa = 0$ at around $r \sim 25$ arcmin. The convergence profiles continue to increase smoothly until $r \sim 40$ arcmin, after this point they start to decrease

and return to the mean background value of $\kappa = 0$. It's clear that the profile of higher redshifts has a larger convergence contrast. Eq.(12) shows that the value of κ is dependent on the line of sight, higher redshift has longer line of sight, which means the absolute magnitude of κ rises. At $r \sim 80$ arcmin some void profiles become slightly underdense, the reason is because the boundary of each void is also the boundary of its neighbour voids, after passing the boundary the profile enters an underdense region again. Fig.(25) (b) is the same profile but the distance to the centroid is rescaled with the effective radius R_V. The crossing point of $\kappa = 0$ is at $r/R_V = 0.7$, the peaks are all around the position $r/R_V = 1$.

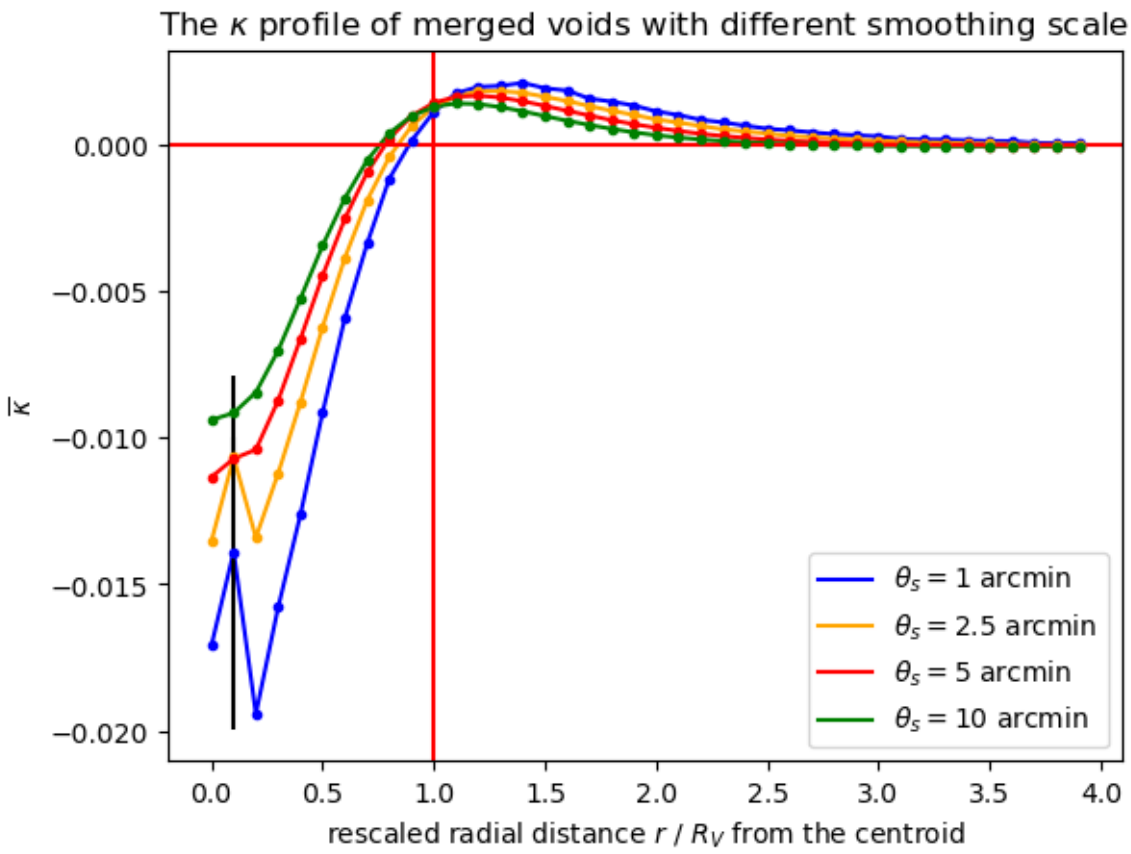
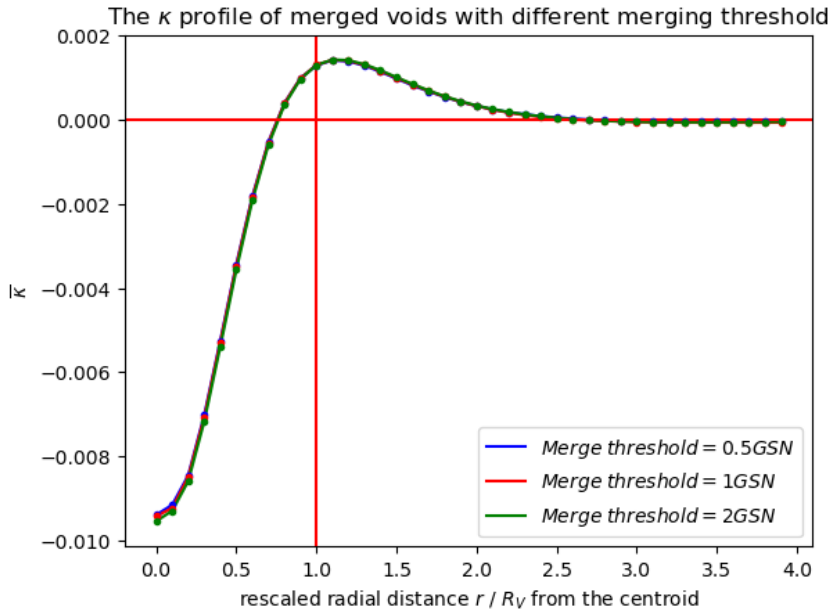
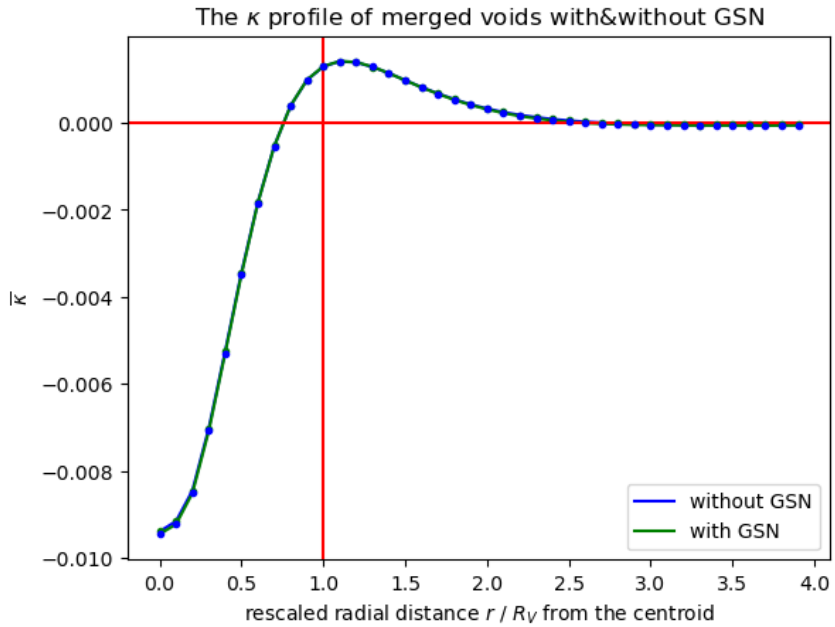


Figure 26: The rescaled convergence profile from void's centroid with different smoothing scale.

An interesting finding is in Fig.(26), it shows the convergence profiles with different smoothing scales, as we can see, the height of the profile peak decreases when the smoothing scale becomes larger, the reason is that the Gaussian filter with larger values erases more convergence information, it can flatten the contrast of convergence perturbations. Another interesting finding is that the peak of the profiles moves from right to $r/R_V = 1$ as the smoothing scale increases, no theoretical explanation can be found currently, we will discuss it in the Chapter 6.



(a)



(b)

Figure 27: The convergence profile from void centroid.

In Fig.(27) (a) we compare the profiles with different merge thresholds, there is no disagreement among these three profiles. This result indicates that the changing of merge thresholds has no impact on the convergence profile. Fig.(27) (b) also shows that the GSN doesn't play an important role in the centroid method convergence profile.

4.3.2 Boundary profiles

Fig.(28) shows the boundary method convergence profiles with five different redshifts. The profiles start at $r \sim$ -50 arcmin, some profiles have noisy shape and the error bars are wide at the beginning, the reason is that as the inward distance from the boundary increases the number of voids decreases, only few voids have a radius larger than 50 arcmin. The peaks of the profiles are located at $r = 0$, after that the profiles sink to become slightly underdense and then back to the mean background value of $\kappa = 0$. The reason for the underdense after peak is that the profile reaches the center region of the neighbouring voids. The boundary profile has the same trend of the curve's steepness and the redshift in Fig.(25), which can be explained by the expression formula of convergence, too. The ends of the profiles also have wide error bars, which is because we calculate the outward profile within range of $2R_{eff}$ from the boundary, the last several data points of $\overline{\kappa}$ correspond to several large voids.

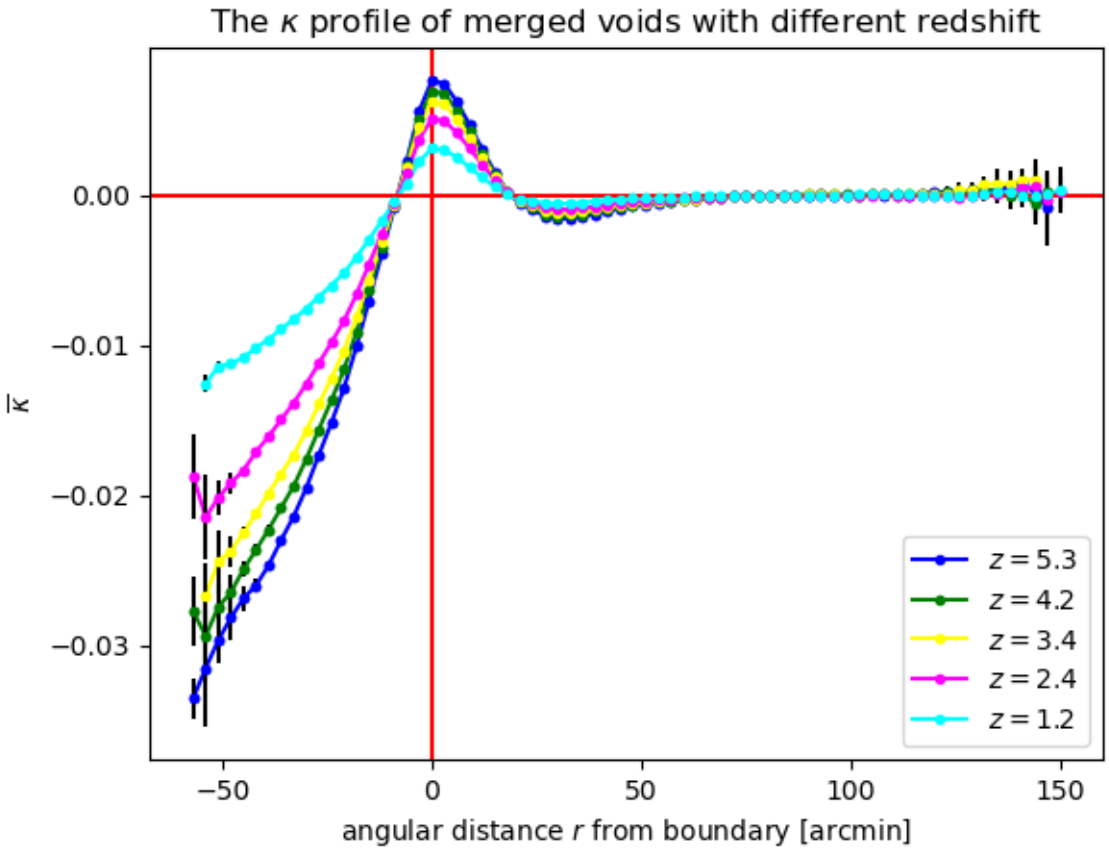


Figure 28: The convergence profile from void's boundary with different red shift.

The impact of the smoothing scale on the boundary method profile is shown in Fig.(29). The starting points for small smoothing scales are closer to $r = 0$ arcmin because the void sizes are smaller for small smoothing scales (as we discussed in Section 4.1). Also, small θ_s has a weaker suppressing effect on the κ perturbation, we can note that the profile of

small θ_s is much steeper than the profile of large θ_s.

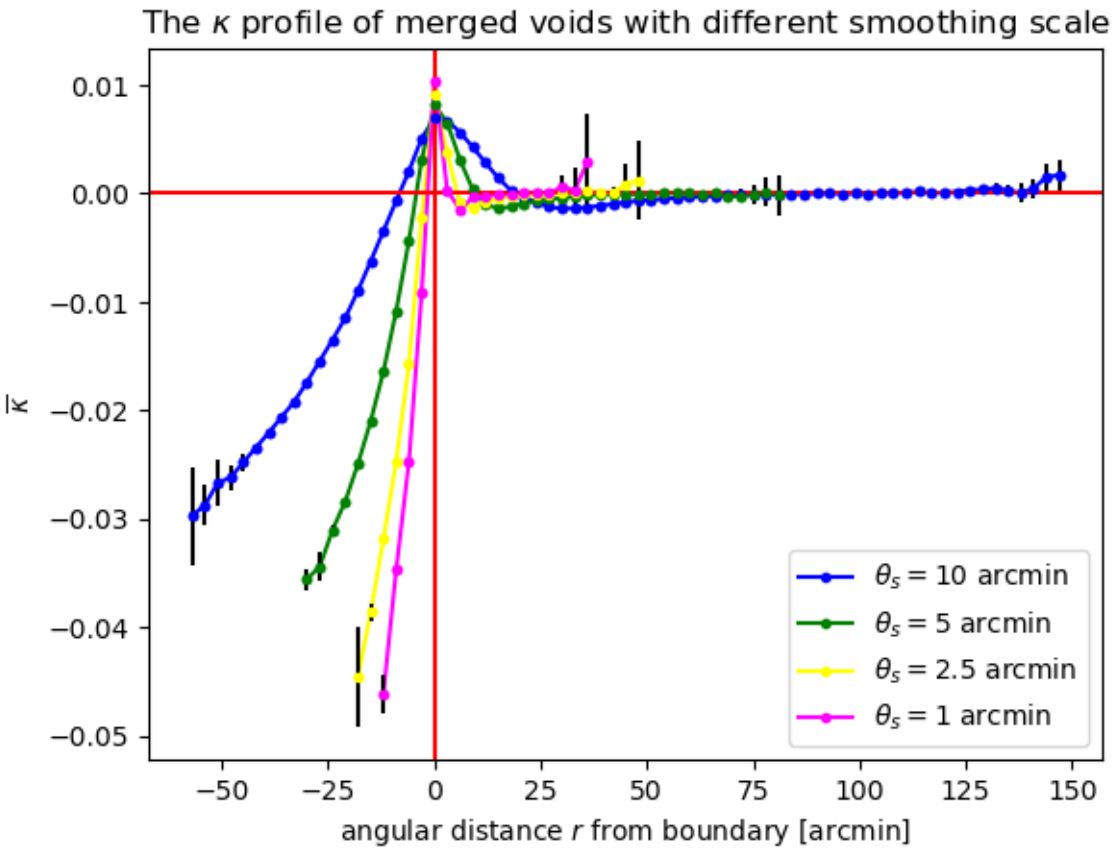
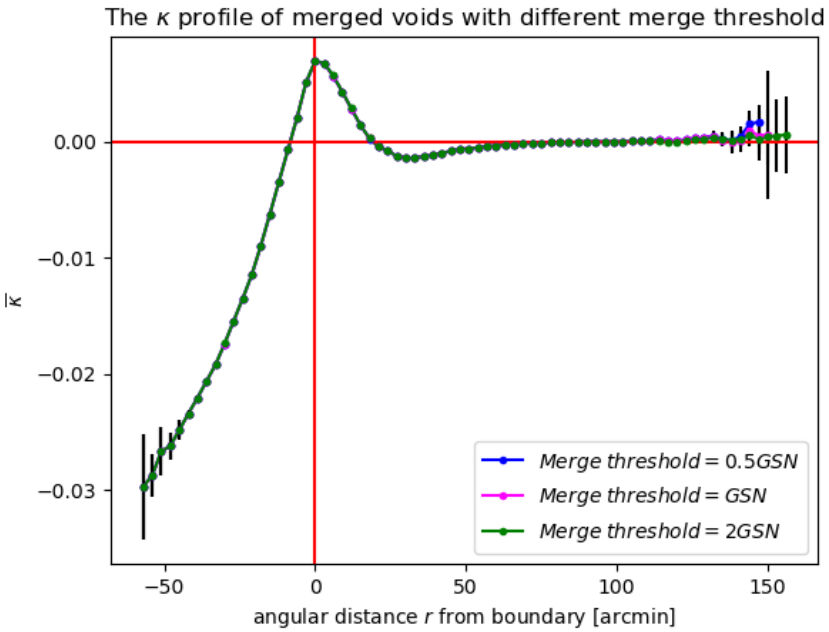
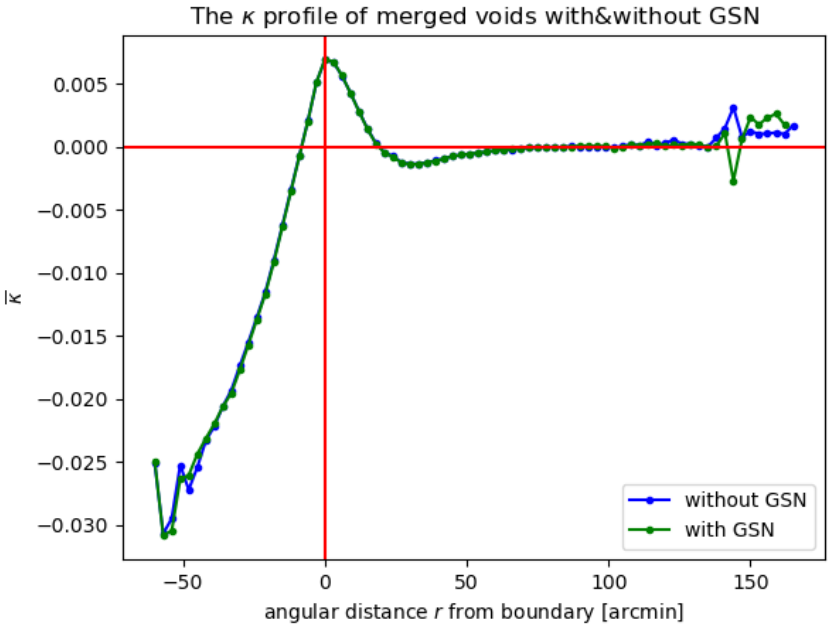


Figure 29: The convergence profile from void's boundary with different smoothing scale.

Fig.(30) (a) shows the profile with different merge thresholds and Fig.(30) (b) shows the profile with and without GSN. The curves overlapped in both figures, which indicates that the merge threshold and GSN have an insignificant effect on the convergence boundary profile.



(a)



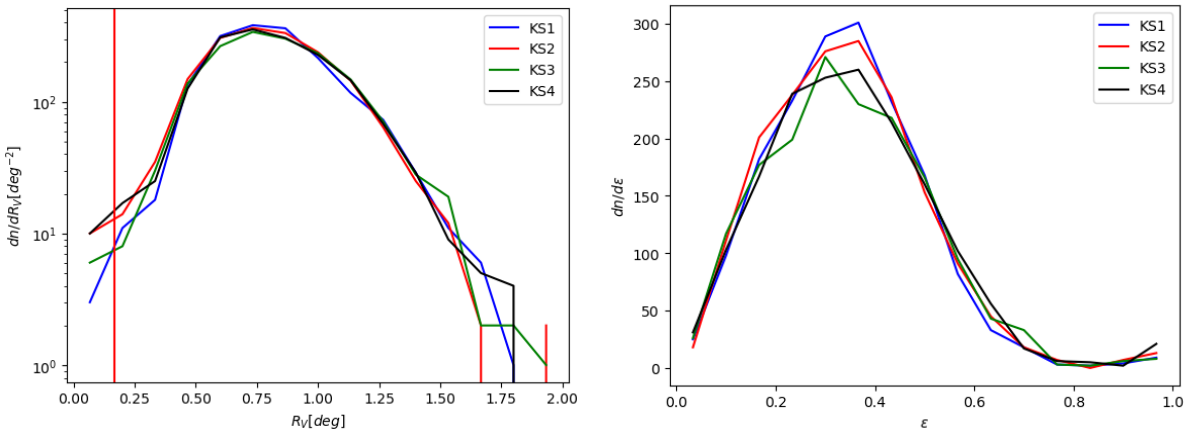
(b)

Figure 30: The convergence profile from void boundary.

5 Results with DES and CosmoGrid

5.1 Result of DES data

Fig.(31) shows the void size abundance and ellipticity distributions of all four tomographic bins, the mask threshold is 10%. The abundances of four tomographic bins in Fig.(31) (a) are all very similar, the peak of these curves is at $R_V \sim 0.75$ deg. The disagreements between the ellipticity distribution in Fig.(31) (b) are still insignificant, the peaks are all located at $\epsilon \sim 0.35$ and the shapes are similar.



(a) (b)

Figure 31: The void size abundance and ellipticity-PDF with different tomographic bins.

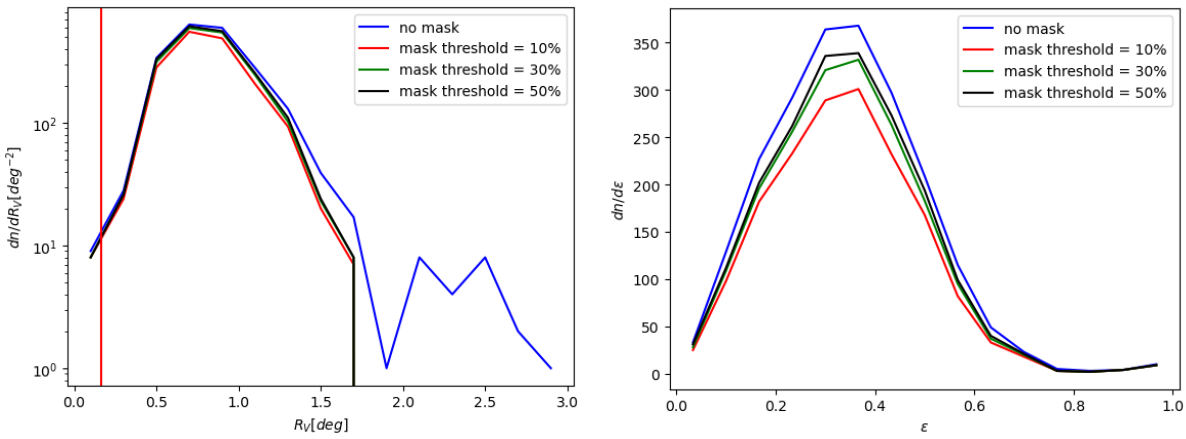
Tab.(3) shows the statistical values of four bins, the number of voids is low and the absolute value of convergence is small compared to the results from the Takahashi simulation in Tab.(1). Despite this, we can still note the same trend for κ, where higher redshift maps have stronger κ perturbations.

Bins	R_V^{max}	R_V^{min}	$\overline{R_V}$	$\overline{\epsilon}$	$\overline{\kappa_{max}}$	$\overline{\kappa_{min}}$	n_V
KS1	1.738	0.097	0.811	0.341	0.004	-0.004	1679
KS2	1.946	0.097	0.805	0.343	0.005	-0.005	1699
KS3	1.892	0.102	0.825	0.346	0.005	-0.005	1594
KS4	1.851	0.107	0.816	0.351	0.006	-0.006	1636

Table 3: The range of the effective radius (R_V^{max} and R_V^{min}), the mean effective radius ($\overline{R_V}$), the mean ellipticity ($\overline{\epsilon}$), the mean maximal and minima convergence ($\overline{\kappa_{max}}$ and $\overline{\kappa_{min}}$), and the mean number of identified voids ($\overline{n_V}$).

The effect varying the mask threshold is shown in Fig.(32). In the left figure we can see several uncommon large voids in the right part of the void abundance without a mask

threshold, these are the voids that covered the vast empty region in Fig.(13) (a). After the mask map applied is to the original results, even the high mask threshold 50% can suppress these fake voids well, there are no significant shape differences among the mask thresholds 10%, 30% and 50%. Unlike the abundance, the mask threshold has no effect on the ellipticity distribution, peaks of the distributions are at same places, only the number of voids is reduced.

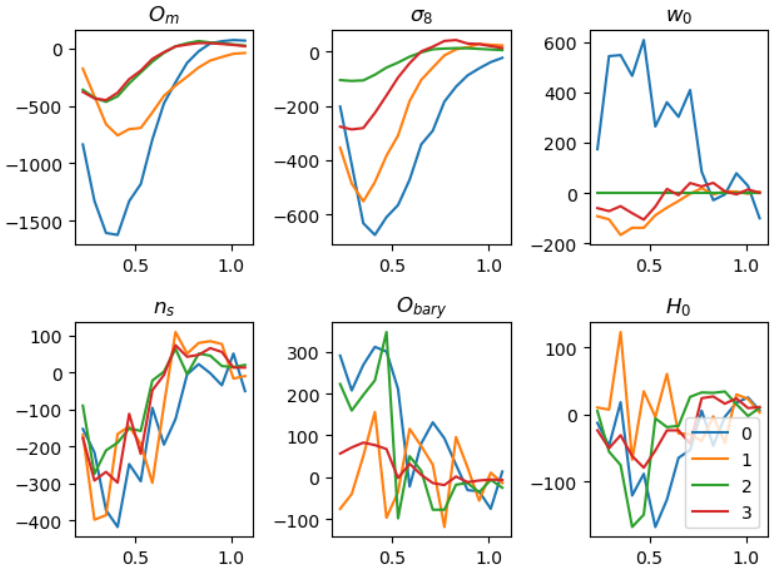


(a) (b)

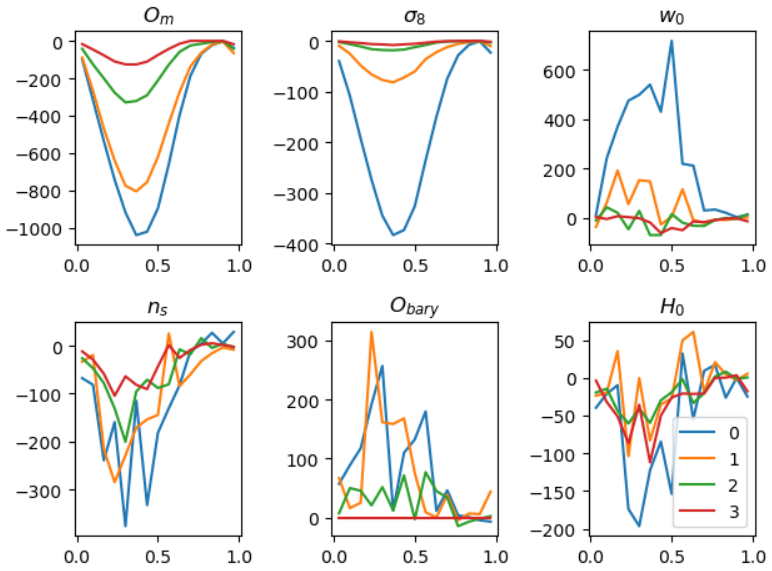
Figure 32: The void size abundance and ellipticity-PDF of the first tomographic bin with different cut-off thresholds.

5.2 Parameter constraints with CosmoGridV1

To check the parameter-sensibility of emulators, we calculate the difference between the results of the maxima and the minimum value in single parameter range (other 5 parameters are assigned with same values of the fiducial cosmology). In Fig.(33) we can see the training effects of 8 emulators. The numbers 0, 1, 2, 3 in legend correspond to the redshift rising tomographic bins. Good sensibility for Ω_m and σ_8 can be observed in the figure, the emulator of the lowest redshift has the most significant difference, the sensibility decrease as the redshift increase.



(a) Emulators of size distribution

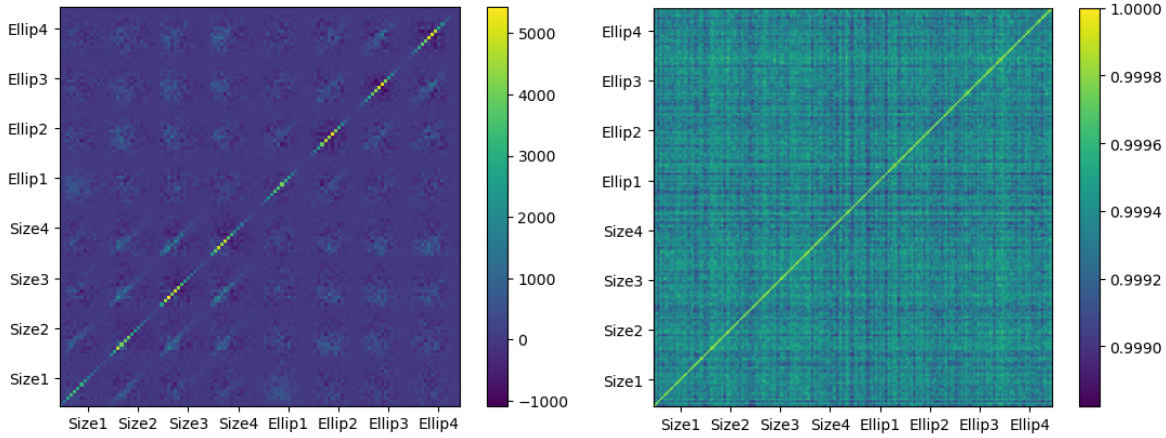


(b) Emulators of ellipticity distribution

Figure 33: The training effects of emulators.

Fig.(35) shows the likelihood contours for measuring the six cosmological parameters with the WL void size and ellipticity distributions. The diagonal panels show the 1D marginalised probability distribution, and remaining panels show the marginalised 2D probability contours enclosing the 68% and 95% confidence intervals. The tightest contours are for the $\sigma_8 - \Omega_m$ plane, which is expected because in WL analysis these are the most sensitive cosmological parameters. For the rest parameters combinations, the contours are very lose, also the 1D marginalised probability distributions are non-Gaussian.

Among all three contours, the contour of ellipticity is larger than other two, which means the constrain power of ellipticity is the weakest. In all parameters combinations panels the combined contours are equal or smaller than other individual contours, though the accuracy is still low. It indicates that the information contained in ellipticity and size distributions are different.



(a) The covariance matrix of all 8 data sets. (b) The correlation matrix of all 8 data sets.

Figure 34: The covariance and correlation matrix of all 8 data sets.

Fig.(34) shows the covariance and correlation matrix of all 8 data vectors. The covariance matrix in Fig.(34) (a) is a 8×8 grid, with rows and columns labeled as size and ellipticity distributions of different tomographic bins. The diagonal elements of the matrix are the variance of each bin in the data vector and the off diagonal elements are the covariance between all possible pairs of bins. The "Size1" to "Size4" data vectors show high positive covariances with each other, indicated by the yellow and green blocks along their intersection on the diagonal. The data vectors "Ellip1" to "Ellip4" also show high positive covariances among themselves. These blocks indicate strong intra-statistic covariances for both "Size" and "Ellip" data vectors. The off diagonal interactions between "Size" and "Ellip" data vectors are less pronounced, suggesting weaker inter-statistic covariances. Fig.(34) (b) shows how much mutual information between different data vectors in the off diagonal terms. The data vectors "Size" and "Ellip" are very highly correlated with each other. It explains why there is no significant improvement when the data vectors are combined.

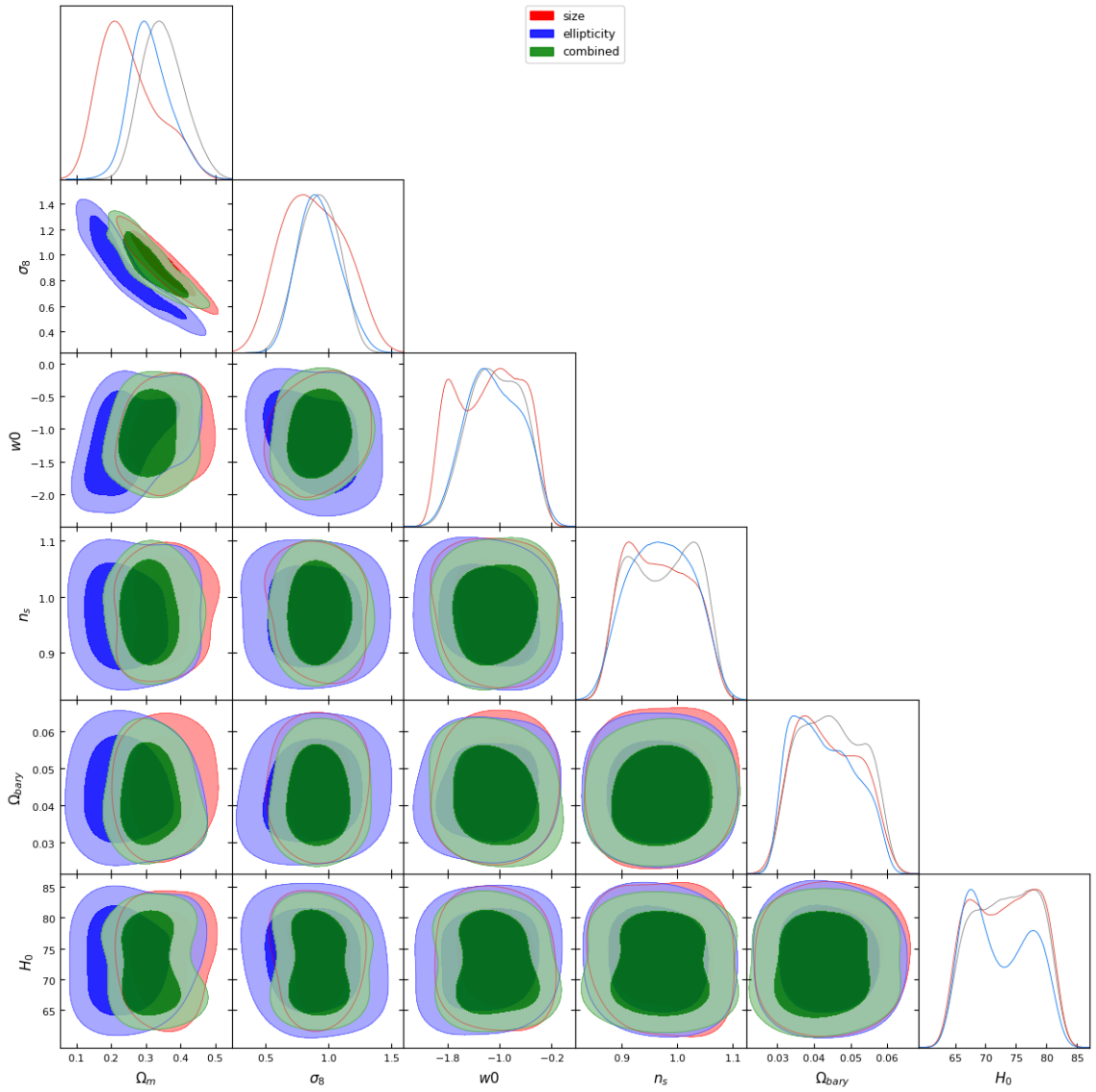


Figure 35: Constraint forecasts on cosmological parameters measured from void size and ellipticity distributions. The diagonal panels show the 1D marginalised probability distribution, and remaining panels show the marginalised 2D probability contours enclosing the 68% and 95% confidence intervals.

6 Discussion and Conclusion

We have presented and discussed the capabilities of **2DWVF**, a new Python algorithm for WL voids finding in cosmological simulations and gravitational lensing surveys. All statistics of 2D watershed voids show the qualitative consistency with previous works. The size and ellipticity distributions have same shapes like 3D watershed voids which identified by **VIDE**. The convergence profile has good agreement with the matter density profile of 3D voids.

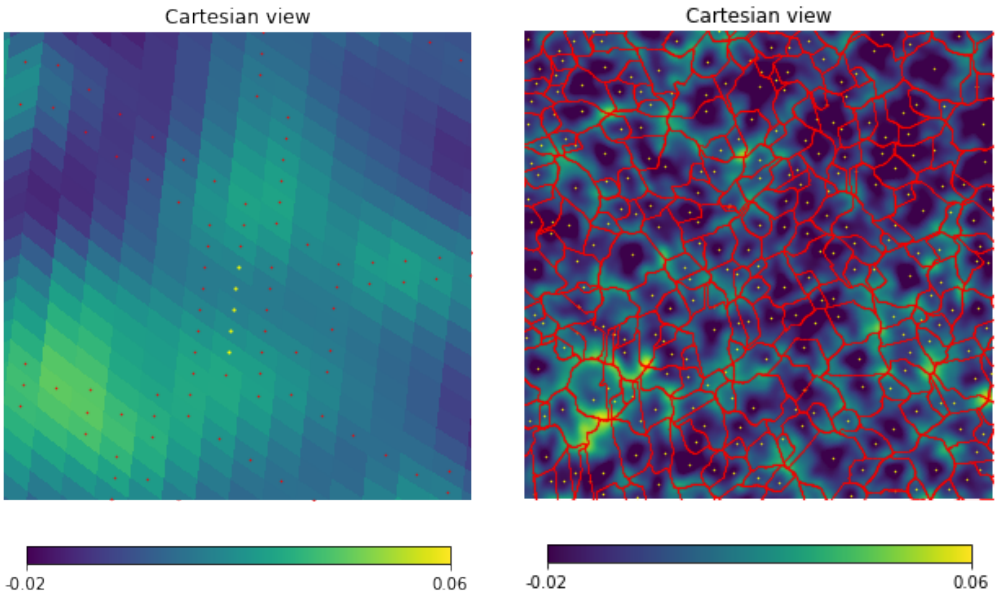
6.1 The void statistics

We have investigated the impact of redshift, smoothing scale, GSN and merge threshold on the statistics of 2D WL voids. We notice the small tails in the most right part of ellipticity distribution, which corresponds to voids with long and narrow shape. In Fig.(17) and Fig.(18) we can find that nearly all extremely abundant high-elliptical voids are small voids, the number of such voids decreases as the smoothing scale rises. We checked the voids with ellipticity larger than 0.95 in the results, most have 1-3 pixel and the largest void only has 5 pixel (see Fig.(36) (a)). Besides, we also found few big voids connect with small tails. In Fig.(36) (b) we can see a void with a tail and several stick-shaped voids in lower left. Similar trend can also be found in the work of Schuster (2023), the voids have been identified with the watershed-based algorithm **VIDE**. Possible reason are the shortcomings of watershed algorithms. The major one is the over-segmentation, which leads to inaccurate segmentation results. Another limitation is that the algorithm is sensitive to noise, because in one loop only one single pixel will be dealt with and it depends on previous, large noise perturbation can further impact the accuracy of the segmentation. The discussion above indicates that though the **2DWVF** can provide more information (ellipticity, boundary convergence profile) than other void finders, because of the high sensibility of noise, it's necessary to find a balanced smoothing scale in the application.

We are not able to find an explanation for the trend observed in Fig.(26). Because Tab.(1) shows that smaller smoothing scale corresponds smaller mean radius, the peak-redshift relation can be considered as peak-mean radius relation. The same trend also occur in the density profile of 3D voids. In the work of Hamaus, Sutter and Wandelt (2014) and Cautun et al. (2016), as the mean void radius increases, the peak of density profile moves to $r/\overline{r_V} = 1$, this trend has been proved true in real observation too (Nadathur et al., 2015). It will be interesting to develop an understanding for this trend in the future.

6.2 The application with real data and cosmology constraints

With the method of mask threshold, our algorithm has been proved applicable for real data. But due to the survey region is limited by the disk of the Milky Way and other bright stars, the map only covers 1/6 sky. We could find around 1600 voids for each tomographic bin, it's much less than the number of voids which have been found in simulation maps, which leads to relatively noisy distributions.



(a) The biggest high-elliptical void in a map.

(b) The segmented voids.

Figure 36: The stick-shaped voids.

The application with CosmoGridV1 shows that 2D watershed voids can be a promising cosmological probe to constrain models. The most sensitive cosmological parameters to the WL voids are σ_8 and Ω_m. Though the constrain power is not high because of high smoothing scale we chose. Also we only used size and ellipticity distributions for the training of emulator and MCMC process. If we expand the training data with other statistics, the constraint power may have an improvement.

6.3 Future work

In the future, there are several possible follow-up studies would directly improve the outcomes of this work.

First, the void statistics haven't been made completely yet. We can make statistics respecting to different mean radius, which can reveal the relationship between void physical properties and sizes. Through the tangential shear profile we can extract more useful information about how images are distorted by cosmic voids in shape. We only applied **2DWVF** with Takahashi simulation, DES and CosmogridV1 in this work. Many convergence maps of simulations and observations can be tested in the future.

Moreover we can calculate the auto- and cross-power spectrum of galaxies, halos and voids on WL maps, these power spectrum can help us to understand the projected matter distribution in the Universe (Hamaus, Wandelt, Sutter, Lavaux and Warren, 2014).

More kinds of statistics can be used for the cosmology constraints. Also, for the data of ellipticity distribution a cut-off threshold can be set to filter high-elliptical voids, in order to minimize the noise because of flaws of the watershed algorithms. It will also be interesting to understand how the WL void function be effected by the intrinsic alignments, baryonic feedback and masking.

Another important work to do is the solution for the over-segmentation of the watershed algorithm. If this shortcoming can be overcome, the **2DWVF** will become the most powerful void finder for WL voids.

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I was a guy who couldn't live without meaning. After some torturous in my early twenties, I thought there could exist some kind of relative meaning rather than absolute meaning. So I defined the meaning of my life as to understand this world. That's why I chose physics, especially astrophysics. Thanks to the academic resources in Munich, I studied the newest cosmology theories and researched the largest objects in this Universe. But after a long way, I found that what we can do is not understand this world but try to understand it. We know that there might be a Big Bang where the evolution of the Universe began, there might be an Inflation epoch which made the formation of LSS possible, there might be something called Dark Energy which leads to the accelerated expansion of the Universe and the explanation of Dark Energy might be Cosmo-

logical constant or Quintessence... But after all, these are not Bibles, these are just the most successful models until now. People use physics models to explain the phenomena but there are always new findings and disagreements. As an agnostist, I doubt that people could ever fully understand this Universe, but that is exactly why this Universe is charming. Compared to the direct way to "success" the possibilities on the halfway are more attractive to me. That's also why the mountains attract me so much. Because there are endless possibilities for the routes, people need to take responsibility for their choices. The summit is not the goal or achievement but the way. The meaning of my life is the way I passed and will pass.

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Declaration

I hereby declare that this thesis is my own work, and that I have not used any sources and aids other than those stated in the thesis.

München, 11. Juni 2024

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Name